

POLARIZATION WITHOUT GROUPS

Anonymous authors

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ABSTRACT

LLM-as-judge systems are increasingly used as scalable substitutes for human annotators, including in polarized, subjective tasks where human annotators systematically disagree along group lines. Whether these systems reproduce, ignore, or distort human polarization patterns is largely untested, in part because existing polarization representations are unreliable. In this work, we find that their unreliability partially rests on the faulty assumption that all polarization can be attributed to at least one annotator group. We also find that these representations can miss polarization entirely when annotators are polarized differently for each item. We posit that these limitations necessitate using a clustering approach, which we use to develop a method that resolves both failure modes, enabling quantifiable attribution of polarization to any annotator group without training. Using this method, we audit LLM-as-judge systems against real human annotation datasets and find that their polarization structure is largely disconnected from human patterns, a gap that persists even when models are explicitly prompted to condition on persona characteristics. We validate the underlying method on simulated data, showing it remains reliable under group-size imbalance and inconsistent polarization effects that degrade current approaches. Overall, our results suggest that persona-conditioning alone is insufficient for LLM judges to replicate the structure of human disagreement.

1 INTRODUCTION

Annotations are essential for subjective tasks, such as content moderation, toxicity, and Hate Speech (HS) detection, which are especially important for protecting minority groups (United Nations, 2025; Dioum, 2025). Human annotation is however unscalable in practical scenarios (e.g., academic research on established datasets, social media deployment), which motivates the use of traditional ML and LLM-as-a-judge models for annotation. Simultaneously, these tasks were traditionally part of safety-tuning new LLMs using RLHF, but because of the cost incurred by manual, human annotation, current practice also incorporates RLAI, where LLM systems are used to train other LLMs (Sharma et al., 2024). The reliance on these models for annotation requires evaluation, which is currently insufficient. Specifically, current practice relies on Inter-Annotator Agreement (IAA) or majority-vote (Ivey et al., 2025; Kralj Novak et al., 2022; van der Velden et al., 2025), which may lead to biased and misconfigured systems (e.g., toxicity models disproportionately marking African American (AA) speech as “toxic” (Sap et al., 2019)), or discard *fundamental* disagreements between minority groups and the majority (Quaranto, 2022), (see the example in Fig. 1).¹

One alternative is using *polarization*, a clustering approach which correlates better with human judgments and is much less sensitive to different group sizes (Pavlopoulos & Likas, 2023; 2024). While we can detect polarization, there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following **Main Research Question (MRQ): To what extent are LLM disagreements consistent with those of human annotators?** We answer this question through the following sub-questions:

RQ 1: Why do current polarization representations fail? While prior work assumed that all polarization can be explained and attributed to some annotator characteristic, through the exhaustive creation of theoretical annotator groups (§4) our study discovers the existence of Inherent Polarization (InPol), which can not be attributed to *any possible group* of annotators—known or unknown

¹The simulated experiments in this study were implemented by generating random annotations following the normal distribution with differing means and Standard Deviation (SD)—see App. G.

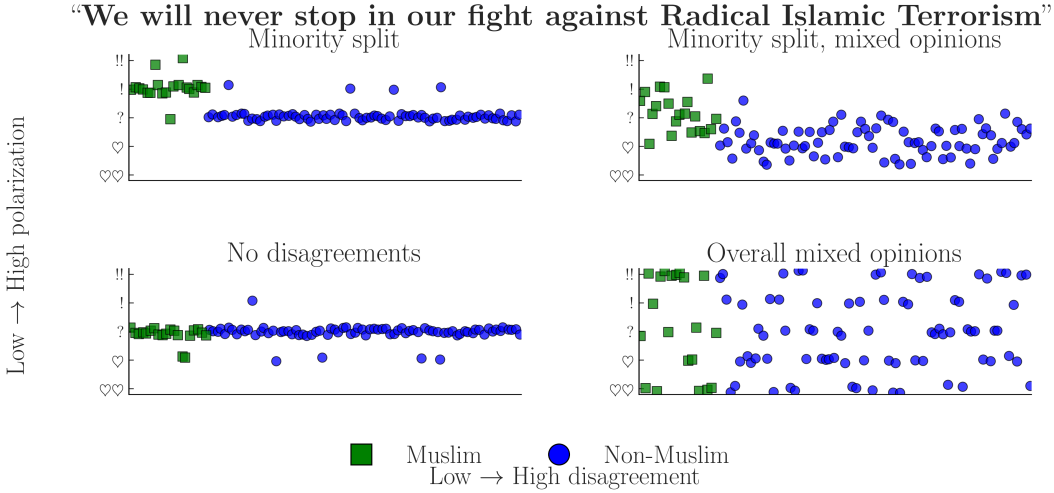


Figure 1: Simulated experiment exploring four scenarios of a HS detection task applied to a controversial statement. The variance in annotations across different annotators may cause traditional IAA metrics to ignore minority opinions, especially when opinions are mixed (top-right).

(Finding 1). We also discover that polarization across items has a *direction*, preventing us from aggregating annotations on the dataset level to circumvent annotation sparsity (Finding 2). We later show that this phenomenon limits existing statistical approaches.

RQ 2: How can we build a reliable representation of polarization attribution? In order to solve these issues we introduce a new metric “*apunim*”, which *quantitatively* attributes polarization to specific Personal Characteristics (PCs) and outperforms prior approaches in scenarios with realistic annotation sparsity and minority group splits (Finding 3). We apply our metric to four Natural Language Processing (NLP) datasets focused on Toxicity and HS detection, confirming prior work showing that annotator gender and ethnicity significantly impact annotation (Finding 4).

RQ 3: Which polarization patterns are shared between human and LLM annotators? We conduct an annotation study on one Toxicity and one HS dataset using six open-source models in order to compare polarization patterns between them and humans (§6). We find that using models using persona-prompting generally fail to produce subgroup polarization patterns regardless of instruction prompt, confirming that persona prompting does not consistently influence LLM outputs in the examined tasks (Finding 5).

Our work makes the following contributions: 1. The identification of severe issues present in subjective datasets and prior methodologies. 2. A new polarization attribution metric that is robust to small annotator counts and minority group splits. 3. An evaluation on how substantial LLM disagreements using persona prompting are in subjective annotation scenarios. We also release a PyPI library implementing our metric.

Content Warning: Controversial and potentially harmful speech used for demonstration.

2 BACKGROUND AND RELATED WORK

Agreement Current work on analyzing subjective tasks rests on using IAA between groups of annotators (Mostafazadeh Davani et al., 2022; Leonardelli et al., 2023; Uma et al., 2022). IAA metrics such as Cohen’s kappa, Krippendorff’s alpha, and Fleiss’s kappa aim to distinguish genuine disagreement from random disagreement through statistical chance-correction. However, these metrics are often difficult to apply to intergroup IAA without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds (Feinstein & Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), producing results that are difficult to compare across studies, being sensitive to class imbalance (Feinstein &

Cicchetti, 1990), and relying on questionable assumptions about chance agreement (Checco et al., 2017). While recent work has addressed some of these limitations with methods that improve applicability and robustness (van der Velden et al., 2025; Wong et al., 2021; Ivey et al., 2025), they are much more complex and less intuitive compared to their traditional counterparts, indicating that IAA may not be the correct tool for analysis of minority viewpoints. Ultimately, IAA analysis may be useful for exploration and identification of some controversial items, but fails at providing an explaining the reason why they are controversial (Abercrombie et al., 2025; Vitsakis et al., 2023).

LLM Annotation Prior work suggests that LLMs are not necessarily reliable annotators when prompted with personas, as they may reproduce stereotypes (Lutz et al., 2025; Liu et al., 2024) and perform unreliably for minority groups (Lutz et al., 2025). To address subjectivity in annotation, several NLP approaches incorporate annotator metadata into LLM fine-tuning (Davani et al., 2023; Röttger et al., 2022; Fleisig et al., 2023; Gupta et al., 2023), although evidence suggests that such improvements may be specific to the datasets on which models are trained (Lee et al., 2023). An alternative line of work recovers annotator perspectives through clustering in embedding space (Vitsakis et al., 2024), although these representations can be difficult to interpret, computationally expensive to obtain and evaluate, and sensitive to the quality of the underlying embeddings. We instead posit that directly clustering the annotations provides a comparatively simple and interpretable mechanism. Our goal differs from studies that evaluate model performance (Gupta et al., 2024) or human-LLM agreement (Dong et al., 2024; Gomez et al., 2024; Masud et al., 2024): as we ask the more limited question of whether LLMs can exhibit attributable polarization, irrespective of whether it corresponds to the annotators’ actual group identities.

Polarization There are multiple competing formulations for polarization besides IAA. One of these approaches (Akhtar et al., 2019) utilizes χ^2 to estimate in-group vs out-group variance. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric (GMM) (Checco et al., 2017), or non-parametric (Pavlopoulos & Likas, 2024; Mignemi et al., 2024) approaches to the annotation histograms. For our work, we use a non-parametric polarization metric, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics (Pavlopoulos & Likas, 2024), although we compare prior approaches in Fig. 4. nDFU (Eq. 1) takes values in the range $[0, 1]$, where values close to 0 indicate no polarization (i.e., opinions are concentrated around a single mode). Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak (see Fig. 2, left for an example).

$$\text{nDFU}(X) = \max_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right) \div f_m, m = \arg \max_i f_i \quad (1)$$

where, X denotes the set of annotations, f_i the relative frequency of the i -th annotation (e.g., the frequency of 2 in a scale of 1–5), and m the histogram peak.

Attributing Polarization to Groups While many studies study the effect of subgroups on IAA (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022), attributing *polarization* is a relatively new concept. Under the name *aposteriori unimodality* (Pavlopoulos & Likas, 2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then it is explained by those groups. Though promising, this mechanism operates on the assumption that *all* polarization can be explained by a single split, which we disprove prove to be impossible in many cases (§4). Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary instead of a quantifiable value.²

3 DATASETS

We use four datasets from current NLP research that include PC information at the level of individual annotators; specifically, the “Kumar” (Kumar et al., 2021) and “Sap” (Sap et al., 2022) datasets,

²For example, when presented with noisy estimates of a metric in the $[0, 1]$ range, we would ideally want to distinguish between cases where attribution is close to 0.001, and cases where attribution is 0.49.

Dataset	#Comments	#Annotators / Item	#PCs	Task
Kumar (Kumar et al., 2021)	2998	5	10	Toxicity
Sap (Sap et al., 2022)	585	5–6	3	Hate Speech
DICES-350 (Aroyo et al., 2023)	350	123	4	Hate Speech
DICES-990 (Aroyo et al., 2023)	990	70–76	4	Hate Speech

Table 1: Datasets used in this study.

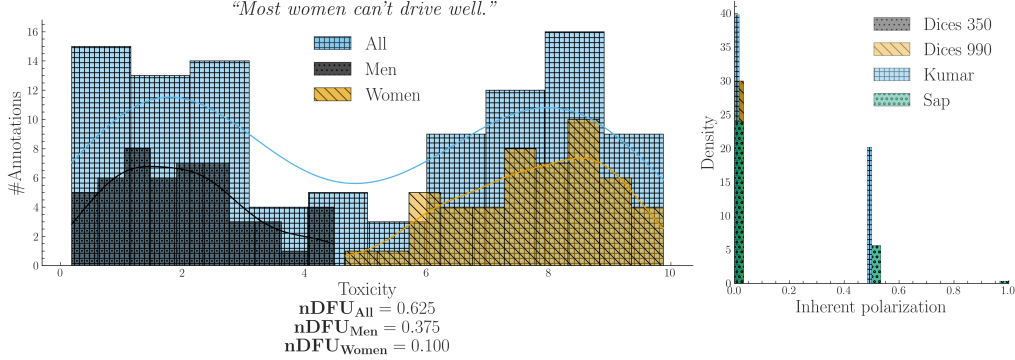


Figure 2: **Left:** Simulated experiment describing a polarizing comment where men and women agree between themselves, but disagree with the opposite gender. **Right:** InPol across our datasets (§3). Contrary to assumptions made in prior work, InPol is commonly non-zero and can even reach $nDFU = 1$ in datasets with low annotator counts.

which are concerned with Toxicity and HS detection respectively. We also examine the DICES-350 and 990 datasets (Aroyo et al., 2023) which focus on LLM response safety.³

A brief overview of dataset characteristics is provided in Table 1. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated items, as well as 10 distinct PC dimensions.⁴ In contrast, the DICES datasets contain fewer annotated items but involve more than an order of magnitude more annotators per item than the other datasets. More details on the annotations, including overall levels of observed polarization, can be found in App. B. For a discussion on the need for datasets with high annotator counts per item, instead of high annotator counts in general, see App. J.

4 CAN WE ATTRIBUTE ALL POLARIZATION TO ANNOTATOR GROUPS?

4.1 PROBLEM FORMULATION

Let $d = \{c_1, c_2, \dots\}$ be a dataset composed of multiple annotated items.⁵ Each item c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \dots\}$, where a_i is a single annotation (e.g., 2 in a LIKERT scale of 1–5), and g_i a group (e.g., men). As an example, the annotations for a item in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\text{"Could be better, could be worse"}) = \{ (+, F), (-, M), (+, F), \dots \}$$

Intuitively, a group partially explains the polarization of a item c when the annotations of that group $A(c | g) = \{a(c, g') \in A(c) | g' = g\}$ exhibit higher polarization compared to the full set of

³Specifically, we use the "Breadth of Posts" dataset with the "toy" labels for Sap. For DICES-350 we choose the *Q3_bias_targeting_inherited_attributes* label, while for DICES-990 the *Q3_bias_incites_hatred* forgoing the *Q3_bias_overall* labels which are aggregated scores. The choice of labels transforms DICES into a HS task with the most coherence since the two datasets do not share labels.

⁴We sample 1,000 items in order to allow for a full LLM annotation later. Our results remain consistent, for both different and differently sized-samples (App. H).

⁵The items could be of any modality (e.g., images, videos, survey questions).

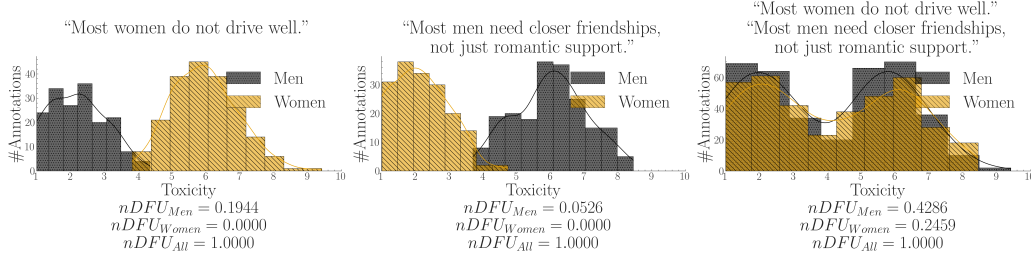


Figure 3: Synthetic experiment showing two comments with annotators disagreeing about which one is toxic. Instead of comparing two unimodal distributions (comments viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions during aggregation, obscuring the source of the polarization (explained by gender, but not shown).

annotations $A(c)$. Consider the case where a misogynistic comment is annotated for toxicity by men and women shown in Fig. 2, left.⁶ The annotations are generally polarized ($nDFU_{all} = 0.62$); but both women ($nDFU_{women} = 0.10$) and men ($nDFU_{men} = 0.37$) exhibit low polarization between their respective groups, suggesting that the overall polarization is partially caused by substantive disagreements between the two examined genders.

4.2 INHERENT POLARIZATION

What Is It? Inherent Polarization (InPol) refers to disagreement that cannot be attributed to *any* arbitrary group. Prior work (Pavlopoulos & Likas, 2024) attributed polarization to groups only when polarization for each individual group became exactly zero (e.g. $nDFU(all) > nDFU(men) = nDFU(women) = 0$). The presence of InPol would invalidate this theory, since there would be no possible group or combination of groups g s.t. $nDFU(g) = 0$. Formally, the InPol for a single item c is the minimum polarization exhibited by any arbitrary group (Eq. 2).

$$Pol_{inh}(c) = \min_{S \subseteq A(c), |S| \geq 3} \{nDFU(S)\} \quad (2)$$

When considered exhaustively, these theoretical, random splits produce both observed and latent groups (e.g., annotators may be less strict if they had lunch prior to annotation). Since nDFU operates only when a group has at least three annotators (Eq. 1), this yields $\sum_{k=3}^n \binom{n}{k}$ possible groups. Since enumerating all subsets is computationally intractable for large annotator counts, we use Monte Carlo sampling by drawing k random partitions with random sizes (Eq. 3) which serves as an upper bound (as it considers a strict subset of Eq. 2).

$$Pol_{inh}^{MC}(c) = \min_{j=1}^k \{nDFU(\tilde{r}_j(A(c)))\} \Rightarrow Pol_{inh}^{MC}(c) \geq Pol_{inh}(c) \forall c \quad (3)$$

where $\tilde{r}$ is an operator that selects a randomly sized, randomly selected subset.

Does It Exist? We compute the InPol for each item across the four datasets. For Sap and Kumar, we use the exhaustive formulation (Eq. 2), whereas for the two DICES datasets we rely on the Monte Carlo approximation with $k = 1000$ (Eq. 3). Fig. 2, right shows that in all DICES items there exists at least one group that fully accounts for the observed polarization even when repeatedly sub-sampling down to 6 annotators.⁷⁸ Interestingly, this observation does not hold for the Kumar and Sap datasets, despite evaluating all possible annotator groups exhaustively. See App. B; Table 9 for a qualitative look through items with high InPol.

Finding 1. *Items may exhibit a degree of InPol that cannot be attributed to any known or latent annotator grouping given the annotators present in the dataset.*

⁶This example can be generalized to k groups; e.g., non-binary or transgender annotators.

⁷ $Pol_{inh}^{MC}(c) = 0 \Rightarrow Pol_{inh}(c) = 0$ due to Eq. 3, as is the case of increased k in Eq. 3.

⁸See App. H for why this observation could be explained by DICES being structurally different.

Can We Remove It? Finding the reason behind InPol is difficult given the lack of datasets (see App. F for some hypotheses and further analysis). If we assume that InPol is caused by a lack of annotations per item, we could address the issue by aggregating annotations between items. However, Fig. 3 shows that two items polarized for the opposite reason may lead to a false negative. Therefore, any statistics must never be aggregated across items.⁹

Finding 2. *Items hold their own pol. direction which may compromise cross-item aggregation.*

5 HOW DO WE ATTRIBUTE THE REST?

5.1 QUANTIFYING ATTRIBUTION TO ANNOTATOR GROUPS

Instead of using InPol, which represents the minimum possible polarization exhibited by any group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups*. Specifically, we define apriori polarization for an item c according to Eq. 4, which is averaged across items for a dataset d (Eq. 5). Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by any single group.

$$Pol_{apr}(c) = \frac{1}{t} \sum_{i=1}^t nDFU(\tilde{r}_i(A(c))) \quad (4) \quad Pol_{apr}(d) = \frac{1}{|d|} \sum_{c \in d} Pol_{apr}(c) \quad (5)$$

where $\tilde{r}$ is the random partition operator (see Eq. 2). We explain why defining apriori polarization in this way is preferable to alternatives in App. E.

We compare this quantity with the average observed polarization of all items (Eq. 6), which we use to compare the polarization that is attributed to a single annotator group with the expected one (Eq. 5), yielding our normalized metric, *apunim* (Eq. 7).

$$Pol_{obs}(d, g) = \frac{1}{|d|} \sum_{c \in d} nDFU(A(c|g)) \quad (6) \quad apunim(d, g) = \frac{Pol_{apr}(d) - Pol_{obs}(d, g)}{1 - Pol_{apr}(d)} \quad (7)$$

Apunim measures the degree to which an annotator group contributes to overall polarization in a set of items. The *sign* of apunim carries information agreement-based attribution does not:

- $apunim(g) \approx 0$: The group g does not meaningfully disagree with the other annotators.
- $apunim(g) > 0$: The group g has fundamental disagreements with the other annotators.
- $apunim(g) < 0$: The annotators of group g are polarized between themselves.

We also include a perturbation-based p-value estimation, which tests whether the observed apunim value of a group g statistically surpasses that of random groups in App. C.

5.2 COMPARISON WITH OTHER METHODS

We compare the effectiveness of apunim, the aposteriori unimodality framework proposed by Pavlopoulos & Likas (2024), the χ^2 method presented in Akhtar et al. (2019), GMM clustering used in Checco et al. (2017), and a baseline using intergroup disagreement ($\Delta\alpha$ Krippendorff) on simulated distributions. To determine whether a score is statistically significant, we construct a null distribution for each method with the null hypothesis that the group labels are unrelated to the observed annotations. We therefore repeatedly shuffle the group labels across annotators, recompute the method’s statistic, and use the resulting scores to form the null distribution. The observed score is then compared with this distribution. See App. C for the apunim p-value implementation.

⁹A similar assumption was followed in early parametric approaches, which assumed different parameters for each item (Checco et al., 2017).

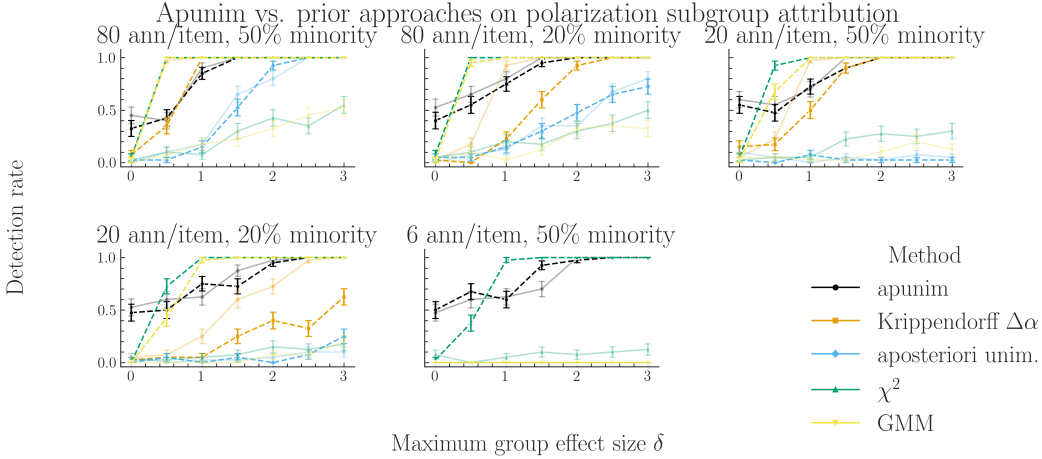


Figure 4: Polarization group attribution on synthetic distributions. Bars indicate standard error. We assign a number of annotators per item, and split the annotator groups 50-50 or 80-20. Higher δ represents higher polarization. Apunim is adept at attributing polarization to the correct group regardless of number of annotators per item, or how these annotators are split. Dashed lines show the results of a simplified polarization setup, where polarization always has the same direction.

Annotations are generated according to $y_{ij} = b_j + g_i p_j + \epsilon_{ij}$, where b_j is the baseline rating for item j , $g_i \in \{-\frac{1}{2}, +\frac{1}{2}\}$ indicates the annotator’s group, p_j controls the magnitude and direction of polarization for item j , and ϵ_{ij} is annotation noise. We try two simulation setups that differ only in how p_j is defined: 1. **Unidirectional polarization**: $p_j \sim \text{Uniform}(0, \delta)$. The group effect always favours the same group across all items, though its magnitude varies per item. 2. **Bidirectional polarization**: $p_j \sim \text{Uniform}(-\delta, +\delta)$. The direction of the group effect can vary across items, so that groups may systematically disagree in opposite directions on different items (see Finding 2).

As shown in Fig. 4, both the χ^2 and GMM methods generally fail at attributing polarization. IAA remains competitive when many annotators are available, but underperforms in cases where the examined group is a minority and when a few annotators are available per item (a common case among NLP datasets—see §3). Apunim in contrast is robust both to the number of annotators per item, opposing polarization effects, and minority groups, validating our initial motivation.

We also observe that the simple statistical approaches do indeed show excellent performance when polarization is unidirectional. This is expected, since they rely on consistent fixed means effects, which is disrupted by opposing polarization effects (see Fig. 3) violating their assumptions on fixed mean differences between groups. Thus, while these methods perform well in ideal conditions, they may completely fail on realistic annotation datasets. While we consider two groups in this experiment, ablations with $k = \{3, 5, 10\}$ groups can be found in App. H; Fig. 7.

Finding 3. *Apunim beats established and proposed methodologies in realistic conditions.*

5.3 HOW ARE HUMANS POLARIZED?

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can explain observed polarization. Statistically significant results are presented in Table 2.¹⁰ Note that the lack of rich annotation data prevents us from analyzing how different annotator groups interact and thus our analysis is necessarily uni-dimensional. See App. D for details on other parameters and execution times, and App. K for the full results.

Polarization is consistently attributed to ethnicity across both DICES datasets, confirming prior work suggesting that it is a systematic source of annotator disagreement in HS tasks (Goyal et al.,

¹⁰The results are not necessarily contingent on a reliable p-value estimation, as non-statistically significant results have very low (< 0.01) apunim values. A threshold on the apunim values would produce similar tables.

Group	Apunim	Support	Group	Apunim	Support
DICES-990			DICES-350		
Ethnicity=AfricanAm	0.099	5148	Ethnicity=AfricanAm	-0.037	7627
Ethnicity=Latino	-0.063	5877	Ethnicity=Asian	0.045	6838
Ethnicity=Other	0.034	21669	Education=Other	-0.023	1841
Ethnicity=White	-0.065	10166	Sap		
Gender=Man	0.036	28817	Gender=man	0.0907	1439
Gender=Woman	-0.027	29990	Gender=woman	-0.2442	877
Age=GenZ	0.047	12214			
Age=Millennial	-0.007	34433			
Age=GenX+	-0.028	12699			
Education=College	0.006	52727			
Education=High school	-0.050	6619			

Table 2: Statistically significant ($p < 0.05$) apunim results across the Sap and DICES datasets. Gray rows show *Ethnicity* and *Gender* groups. “Support” refers to number of annotations. Consult App. K for the full results.

Table 3: Mean inherent polarization (mean $\pm$ 2 SD) per (dataset, prompt), for Human and each LLM.

Dataset	Source	Default	Stereotype	Persona	Single	Direct
kumar	Human	0.17 $\pm$ 0.473	0.17 $\pm$ 0.473	0.17 $\pm$ 0.473	0.17 $\pm$ 0.473	0.17 $\pm$ 0.473
	llama70b	0.00 $\pm$ 0.032	0.00 $\pm$ 0.000	0.00 $\pm$ 0.045	0.00 $\pm$ 0.045	0.00 $\pm$ 0.045
	olmo32b	0.01 $\pm$ 0.137	0.03 $\pm$ 0.236	0.02 $\pm$ 0.203	0.03 $\pm$ 0.248	0.02 $\pm$ 0.201
	qwen32b	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.032	0.00 $\pm$ 0.000
	qwen7b	0.00 $\pm$ 0.095	0.01 $\pm$ 0.137	0.00 $\pm$ 0.083	0.00 $\pm$ 0.095	0.01 $\pm$ 0.109
sap	Human	0.11 $\pm$ 0.437	0.11 $\pm$ 0.437	0.11 $\pm$ 0.437	0.11 $\pm$ 0.437	0.11 $\pm$ 0.437
	llama70b	0.00 $\pm$ 0.058	0.00 $\pm$ 0.041	0.00 $\pm$ 0.058	0.00 $\pm$ 0.000	0.00 $\pm$ 0.041
	olmo32b	0.01 $\pm$ 0.136	0.01 $\pm$ 0.142	0.01 $\pm$ 0.109	0.01 $\pm$ 0.163	0.01 $\pm$ 0.148
	qwen32b	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.000	0.00 $\pm$ 0.041	0.00 $\pm$ 0.000
	qwen7b	0.00 $\pm$ 0.000	0.00 $\pm$ 0.071	0.00 $\pm$ 0.041	0.00 $\pm$ 0.041	0.00 $\pm$ 0.041

2022).¹¹ Additionally, in both DICES-990 and Sap, *women* are a consistent source of polarization (positive values), while *men* disagree among themselves (negative values), which is consistent with previous findings (Wojatzki et al., 2018; Binns et al., 2017). None of the Kumar dimensions show statistically significant values, but this is expected, as our methodology requires at least six annotators per item to work reliably.

Finding 4. *Annotator ethnicity and gender are consistent causes of polarization, confirming prior literature on Toxicity and HS tasks.*

6 WHAT POLARIZATION PATTERNS DO LLM ANNOTATORS EXHIBIT?

Experimental Procedure We investigate four¹² open-source LLMs with six annotators per item¹³ supplied with personas sampled from our human dataset. We test polarization structure on three fronts: 1. Observed polarization (nDFU). 2. Overall InPol. 3. Apunim values for each group. See App. I for the ablation experimental procedure, full results, model details, dataset selection, exploratory analysis, and comparison to human annotations.

¹¹Note that polarization attribution does not have a consistent direction across datasets, which we believe to be caused by studies using different ethnic subgroups allowing detection but distorting attribution.

¹²We used six models, but discarded two since they proved unreliable—see ablations in App. I.

¹³It is possible that LLMs can produce polarized patterns when scaled up to dozens or hundreds of annotators. This however is both hard to check due to exorbitant computational costs (App. I) and hard to scale up in realistic deployment scenarios.

Table 4: Cohen’s d statistics showing the quantitative difference in nDFU between the default and each adversarial instruction prompt for each of the groups of both datasets and across all models.

Model		Mean	Std	Min	25%	50%	75%	Max
llama70b	stereotype	0.057	0.067	-0.059	0.021	0.036	0.066	0.227
	persona	0.013	0.090	-0.102	-0.038	-0.020	0.008	0.255
	single	0.167	0.041	0.084	0.137	0.159	0.185	0.254
	direct	-0.062	0.093	-0.172	-0.113	-0.095	-0.062	0.182
olmo32b	stereotype	0.022	0.031	-0.058	0.001	0.021	0.034	0.144
	persona	0.154	0.051	0.018	0.153	0.172	0.182	0.288
	single	0.311	0.075	0.215	0.272	0.285	0.311	0.505
	direct	0.122	0.098	-0.132	0.135	0.156	0.170	0.236
qwen32b	stereotype	0.004	0.041	-0.088	-0.017	0.005	0.031	0.111
	persona	0.104	0.065	-0.073	0.098	0.122	0.138	0.216
	single	0.121	0.081	-0.091	0.124	0.149	0.165	0.241
	direct	0.092	0.062	-0.077	0.079	0.105	0.125	0.204
qwen7b	stereotype	-0.052	0.075	-0.172	-0.096	-0.073	-0.040	0.161
	persona	0.106	0.056	0.023	0.069	0.088	0.122	0.258
	single	-0.047	0.066	-0.199	-0.090	-0.063	-0.025	0.128
	direct	-0.120	0.090	-0.230	-0.170	-0.154	-0.107	0.124

Are LLMs Polarized by Default? We first examine whether persona prompting with standard instructions leads to polarized annotations. Our results show *no statistically significant attribution effects* and very small *apunim* values for any of the groups in all models and datasets (App. I; Table 16, 17). Unlike human annotations, we observe no InPol (Table 3) and only two of the four models showing any polarization at all (App. I; Fig. 10).

Do Instructions Influence Polarization? Since instruction prompts are known to influence the effect of personas (Weeber et al., 2026), we also try four adversarial prompts progressively emphasizing the importance of the LLMs’ personas (App. L). While an ANOVA test¹⁴ shows that the choice of instruction prompt does indeed affect polarization across most models (almost all groups for OLMo-32B, half for LLaMa-70B, Qwen-7B and a single group for Qwen-32B), these effects are almost always non-qualitatively significant (< 0.2 Cohen’s d), as shown in Table 4. The “single” prompt is the only prompt producing a sometimes notable (small-to-medium) quantitative effect exclusively on the OLMo-32B model, which is indeed significant as evidenced by App. I; Fig. 10. Our findings support social science work which warns against generating human annotation data without substantial replication studies (Rossi et al., 2024; Kapania et al., 2025).

Finding 5. *LLMs with persona-prompting do not produce annotations with detectable InPol, nor with attributable polarization across any group, while only a few exhibit any polarization at all. This effect can not be reliably remedied with prompt engineering.*

CONCLUSION

We discovered issues regarding the presence of inherent and conflicting polarization which render current representations of subgroup disagreement attribution suboptimal in realistic circumstances. We created a new metric that overcomes these issues and outperforms alternative methodologies in realistic annotation settings, while also quantifying the attribution of each group. By applying this metric to two subjective NLP datasets, we verify that ethnicity and gender seem to generally drive polarization attribution in Toxicity and HS tasks. Finally, we find that persona prompting is not generally capable of unlocking consistent subgroup disagreement even when using adversarial instruction prompts.

¹⁴We use a parametric test due to the abundance of observations ($N = 585, 1000$). We also use a multiple hypothesis p-value correction (specifically the “Holms-Bonferroni” method (Holm, 1979)).

AI USE STATEMENT

In this work we used proprietary models such as Claude’s Sonnet 5 and OpenAI’s GPT-5 for partially implementing the synthetic experiments in §5.2 and generating parts of the library, experiment and graph-creation code. The unit-tests used in our library were similarly generated by these models, although new tests were added as edge-cases and bugs were discovered. All such tasks were executed with direct author supervision, and all relevant code files (library, experiments, graphs) have been edited manually at multiple points by the authors to ensure reproducibility, accuracy and software quality. We also used Scholar Labs to find citations for targeted claims (e.g., for known issues with persona prompting), although all such citations were manually filtered by us. Finally, we used a locally hosted Gemma-4 model (specifically the `gemma-4-E4B-it-Q4_K_M.gguf` variant) for rephrasing this manuscript in order to improve readability. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

We ensure that our LLM annotation study is reproducible by selecting exclusively open-source, locally hosted models and fixing all relevant seeds. An ablation study where we repeat annotations multiple times shows that our setup achieves 100% consistency across all models. In order to ensure that the LLM annotators are not influenced by minute differences in the prompts, we execute a second ablation study where we rephrase the main annotation prompt three times and compare outputs. This procedure revealed that two models were especially sensitive to such changes, which led us to exclude those models from the analysis presented in this manuscript. Information on the synthetic experiments used in §4.5 is disclosed in App. G. We analyze software design decisions, limitations and computational details in App. D and elaborate on our experimental setup in App. E, while ablations around sample sizes and sample selection for the Kumar dataset are detailed in App. H. Additionally, we include a discussion around the use of current NLP datasets, as well as an empirical estimation of the number of required annotators for reproducible results in App. J. All code, environment details (e.g., libraries used), data-processing scripts and our library implementing *apunim* are released anonymously: library <https://anonymous.4open.science/r/apunim-5F70>, online (HTML) documentation <https://anonymous.4open.science/r/apunim-page>, replication code <https://anonymous.4open.science/r/aposteriori-unimodality-E8C1>.

ETHICS STATEMENT

Dual Use Concerns The primary objective of this study is to identify systemic sources of polarization in subjective NLP datasets to facilitate the construction of more fair and reliable AI systems. However, given that our methodology allows for the quantitative attribution of specific annotation patterns to PCs, we acknowledge three primary risks inherent in this research. 1. *Discriminatory Profiling*: If the metric is applied outside of a dataset quality analysis, its ability to correlate specific opinion patterns with annotator groups could be misused for surveillance or unjust profiling. 2. *Automated Bias Amplification*: The diagnostic patterns of polarization we detect are reflections of human systemic biases. Without human oversight, training subsequent models on these patterns risks automating and entrenching existing societal prejudices. 3. The knowledge provided by *apunim* could be *weaponized*, allowing malicious actors to strategically target weak points in moderation systems or intentionally maximize societal conflict. We stress that our metric functions strictly as a diagnostic warning signal. Any deployment in a real-world system must mandate a human-in-the-loop review, ensuring that identified patterns lead to systemic improvements (e.g., policy reform, demographic diversity initiatives) rather than merely justifying an automated system. In any case and despite the risks, we believe that the benefits of this research outweigh the risks outlined above.

Data Sourcing and Privacy All datasets employed in this study were either publicly accessible or provided to us by the original authors with full disclosure regarding the purpose of our research. No identifiable, personal information was included in our pipeline. We do not release proprietary data provided by the authors, nor have we provided any to proprietary LLMs. Finally, we note that our

data is limited to English posts from a U.S. cultural context and our results may not be representative for other cultures.

Examples Used Our study utilizes examples of comments that may be considered divisive across certain demographics. These examples were generated by our team solely for demonstration purposes. We explicitly state that the contents of these examples and the groups studied do not necessarily reflect the views or behaviors of the groups they represent, nor do they reflect the opinions of the authors. We have partially censored explicit slurs in Table 9 in order to both mitigate uncomfortable or triggering content, while also showing enough content for the examples to be effective.

Prompts Used This study uses adversarial prompts which explicitly instruct the model to change its outputs according to their persona prompts. The purpose of these prompts is to test the LLMs under an extreme case of prompt engineering and are not to be used for generalized usage. The use of these prompts may degrade model performance, increase the risk of stereotyping and produce misleading results.

REFERENCES

- Gavin Abercrombie, Tanvi Dinkar, Amanda Cercas Curry, Verena Rieser, and Dirk Hovy. Consistency is key: Disentangling label variation in natural language processing with intra-annotator agreement. In Gavin Abercrombie, Valerio Basile, Simona Frenda, Sara Tonelli, and Shiran Dudy (eds.), *Proceedings of the The 4th Workshop on Perspectivist Approaches to NLP*, pp. 63–74, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-350-0. doi: 10.18653/v1/2025.nlperspectives-1.6. URL <https://aclanthology.org/2025.nlperspectives-1.6/>.
- Sohail Akhtar, Valerio Basile, and Viviana Patti. A new measure of polarization in the annotation of hate speech. In Mario Alviano, Gianluigi Greco, and Francesco Scarcello (eds.), *AI*IA 2019 – Advances in Artificial Intelligence*, pp. 588–603, Cham, 2019. Springer International Publishing. ISBN 978-3-030-35166-3.
- Hala Al Kuwatly, Maximilian Wich, and Georg Groh. Identifying and measuring annotator bias based on annotators’ demographic characteristics. In *Proceedings of the fourth workshop on online abuse and harms*, pp. 184–190, 2020.
- Lora Aroyo, Alex Taylor, Mark Díaz, Christopher Homan, Alicia Parrish, Gregory Serapio-García, Vinodkumar Prabhakaran, and Ding Wang. Dices dataset: Diversity in conversational ai evaluation for safety. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 53330–53342. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/a74b697bce4cac6c91896372abaa8863-Paper-Datasets_and_Benchmarks.pdf.
- Reuben Binns, Michael Veale, Max Van Kleek, and Nigel Shadbolt. Like trainer, like bot? inheritance of bias in algorithmic content moderation. In *International conference on social informatics*, pp. 405–415. Springer, 2017.
- James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. Synthetic replacements for human survey data? the perils of large language models. *Political Analysis*, 32(4):401–416, 2024. doi: 10.1017/pan.2024.5.
- Ted Byrt, Janet Bishop, and John B. Carlin. Bias, prevalence and kappa. *Journal of Clinical Epidemiology*, 46(5):423–429, May 1993. ISSN 0895-4356. doi: 10.1016/0895-4356(93)90018-V. URL [https://doi.org/10.1016/0895-4356\(93\)90018-V](https://doi.org/10.1016/0895-4356(93)90018-V).
- Alessandro Checco, A. Roitero, Eddy Maddalena, S. Mizzaro, and Gianluca Demartini. Let’s agree to disagree: Fixing agreement measures for crowdsourcing. 10 2017.
- S.Y. Chen, Z. Feng, and X. Yi. A general introduction to adjustment for multiple comparisons. *Journal of Thoracic Disease*, 9(6):1725–1729, 2017. doi: 10.21037/jtd.2017.05.34; PMID: 28740688; PMCID: PMC5506159.

- Aida Mostafazadeh Davani, Mohammad Atari, Brendan Kennedy, and Morteza Dehghani. Hate speech classifiers learn normative social stereotypes. *Transactions of the Association for Computational Linguistics*, 11:300–319, 2023. doi: 10.1162/tacl_a_00550. URL <https://aclanthology.org/2023.tacl-1.18/>.
- Shelby Dioum. What explains the increase in online hate speech?, 3 2025. URL <https://www.ucdavis.edu/magazine/what-explains-increase-online-hate-speech>. Accessed: 2025-09-22.
- Yijiang River Dong, Tiancheng Hu, and Nigel Collier. Can LLM be a personalized judge? In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2024*, pp. 10126–10141, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.592. URL <https://aclanthology.org/2024.findings-emnlp.592/>.
- European Union. Regulation (EU) 2016/679 of the European Parliament and of the Council, 2016. URL <https://data.europa.eu/eli/reg/2016/679/oj>.
- European Union. Regulation (eu) 2024/1689 of the european parliament and of the council of 13 june 2024 laying down harmonised rules on artificial intelligence (artificial intelligence act). <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>, 2024. URL <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>. Official Journal of the European Union, L 2024/1689.
- Alvan R. Feinstein and Domenic V. Cicchetti. High agreement but low kappa: I. the problems of two paradoxes. *Journal of Clinical Epidemiology*, 43(6):543–549, Jan 1990. ISSN 0895-4356. doi: 10.1016/0895-4356(90)90158-L. URL [https://doi.org/10.1016/0895-4356\(90\)90158-L](https://doi.org/10.1016/0895-4356(90)90158-L).
- Eve Fleisig, Rediet Abebe, and Dan Klein. When the majority is wrong: Modeling annotator disagreement for subjective tasks. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 6715–6726, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.415. URL <https://aclanthology.org/2023.emnlp-main.415/>.
- Tommaso Giorgi, Lorenzo Cima, Tiziano Fagni, Marco Avvenuti, and Stefano Cresci. Human and llm biases in hate speech annotations: A socio-demographic analysis of annotators and targets. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 19, pp. 653–670, 2025.
- Juan Felipe Gomez, Caio Machado, Lucas Monteiro Paes, and Flavio Calmon. Algorithmic arbitrariness in content moderation. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, FAccT ’24, pp. 2234–2253, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400704505. doi: 10.1145/3630106.3659036. URL <https://doi.org/10.1145/3630106.3659036>.
- Nitish Goyal, Ian D Kivlichan, Rachel Rosen, and Lucy Vasserman. Is your toxicity my toxicity? exploring the impact of rater identity on toxicity annotation. *Proceedings of the ACM on Human-Computer Interaction*, 6(CSCW2):1–28, 2022.
- Aaron Grattafiori et al. The llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Dirk Groeneveld et al. OLMo: Accelerating the science of language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 15789–15809, Bangkok, Thailand, aug 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.841. URL <https://aclanthology.org/2024.acl-long.841/>.
- Shashank Gupta, Vaishnavi Shrivastava, Ameet Deshpande, Ashwin Kalyan, Peter Clark, Ashish Sabharwal, and Tushar Khot. Bias runs deep: Implicit reasoning biases in persona-assigned llms. In *International Conference on Learning Representations*, volume 2024, pp. 21849–21874, 2024.

- Soumyajit Gupta, Sooyong Lee, Maria De-Arteaga, and Matthew Lease. Same same, but different: Conditional multi-task learning for demographic-specific toxicity detection. In *Proceedings of the ACM Web Conference 2023*, WWW '23, pp. 3689–3700, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9781450394161. doi: 10.1145/3543507.3583290. URL <https://doi.org/10.1145/3543507.3583290>.
- Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. ISSN 03036898, 14679469. URL <http://www.jstor.org/stable/4615733>.
- Jonathan Ivey, Susan Gauch, and David Jurgens. NUTMEG: Separating signal from noise in annotator disagreement. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (eds.), *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 2874–2887, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.144. URL <https://aclanthology.org/2025.emnlp-main.144/>.
- Shivani Kapania, Stephanie Ballard, Alex Kessler, and Jennifer Wortman Vaughan. Examining the expanding role of synthetic data throughout the ai development pipeline. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*, FAccT '25, pp. 45–60, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400714825. doi: 10.1145/3715275.3732005. URL <https://doi.org/10.1145/3715275.3732005>.
- Petra Kralj Novak, Teresa Scantamburlo, Andraž Pelicon, Matteo Cinelli, Igor Mozetič, and Fabiana Zollo. Handling disagreement in hate speech modelling. In Davide Ciucci, Inés Couso, Jesús Medina, Dominik Ślęzak, Davide Petturiti, Bernadette Bouchon-Meunier, and Ronald R. Yager (eds.), *Information Processing and Management of Uncertainty in Knowledge-Based Systems*, pp. 681–695, Cham, 2022. Springer International Publishing. ISBN 978-3-031-08974-9.
- Deepak Kumar, Patrick Gage Kelley, Sunny Consolvo, Joshua Mason, Elie Bursztein, Zakir Durumeric, Kurt Thomas, and Michael Bailey. Designing toxic content classification for a diversity of perspectives. In *Proceedings of the Seventeenth USENIX Conference on Usable Privacy and Security*, SOUPS'21, USA, 2021. USENIX Association. ISBN 978-1-939133-25-0.
- Noah Lee, Na Min An, and James Thorne. Can large language models capture dissenting human voices? In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 4569–4585, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.278. URL <https://aclanthology.org/2023.emnlp-main.278/>.
- Elisa Leonardelli, Gavin Abercrombie, Dina Almanea, Valerio Basile, Tommaso Fornaciari, Barbara Plank, Verena Rieser, Alexandra Uma, and Massimo Poesio. SemEval-2023 task 11: Learning with disagreements (LeWiDi). In Atul Kr. Ojha, A. Seza Doğruöz, Giovanni Da San Martino, Harish Tayyar Madabushi, Ritesh Kumar, and Elisa Sartori (eds.), *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, pp. 2304–2318, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.semeval-1.314. URL <https://aclanthology.org/2023.semeval-1.314/>.
- Andy Liu, Mona Diab, and Daniel Fried. Evaluating large language model biases in persona-steered generation. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 9832–9850, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.586. URL <https://aclanthology.org/2024.findings-acl.586/>.
- Marlene Lutz, Indira Sen, Georg Ahnert, Elisa Rogers, and Markus Strohmaier. The prompt makes the person(a): A systematic evaluation of sociodemographic persona prompting for large language models. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 23212–23237, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-335-7. doi: 10.18653/v1/2025.findings-emnlp.1261. URL <https://aclanthology.org/2025.findings-emnlp.1261/>.

- Sarah Masud, Sahajpreet Singh, Viktor Hangya, Alexander Fraser, and Tanmoy Chakraborty. Hate personified: Investigating the role of LLMs in content moderation. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 15847–15863, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.886. URL <https://aclanthology.org/2024.emnlp-main.886/>.
- Giuseppe Mignemi, Antonio Calcagni, Andrea Spoto, and Ioanna Manolopoulou. Mixture polarization in inter-rater agreement analysis: a bayesian nonparametric index. *Statistical Methods & Applications*, 33(1):325–355, Mar 2024. ISSN 1613-981X. doi: 10.1007/s10260-023-00741-x. URL <https://doi.org/10.1007/s10260-023-00741-x>.
- Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. Dealing with disagreements: Looking beyond the majority vote in subjective annotations. *Transactions of the Association for Computational Linguistics*, 10:92–110, 2022. doi: 10.1162/tacl_a_00449. URL <https://aclanthology.org/2022.tacl-1.6/>.
- Team Olmo. Olmo 3, 2026. URL <https://arxiv.org/abs/2512.13961>.
- John Pavlopoulos and Aristidis Likas. Distance from unimodality for the assessment of opinion polarization. *Cognitive Computation*, 15(2):731–738, 2023. ISSN 1866-9964. doi: 10.1007/s12559-022-10088-2. URL <https://doi.org/10.1007/s12559-022-10088-2>.
- John Pavlopoulos and Aristidis Likas. Polarized opinion detection improves the detection of toxic language. In Yvette Graham and Matthew Purver (eds.), *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1946–1958, St. Julian’s, Malta, March 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.eacl-long.117/>.
- Anne Quaranto. Dog whistles, covertly coded speech, and the practices that enable them. *Synthese*, 200(4):330, 2022.
- Amirhossein Razavi, Mina Soltangheis, Negar Arabzadeh, Sara Salamat, Morteza Zihayat, and Ebrahim Bagheri. Benchmarking prompt sensitivity in large language models. In Claudia Hauff, Craig Macdonald, Dietmar Jannach, Gabriella Kazai, Franco Maria Nardini, Fabio Pinelli, Fabrizio Silvestri, and Nicola Tonellotto (eds.), *Advances in Information Retrieval*, pp. 303–313, Cham, 2025. Springer Nature Switzerland. ISBN 978-3-031-88714-7.
- Luca Rossi, Katherine Harrison, and Irina Shklovski. The problems of llm-generated data in social science research. *Sociologica*, 18(2):145–168, 1 2024. doi: 10.6092/issn.1971-8853/19576. URL <https://sociologica.unibo.it/article/view/19576>.
- Paul Röttger, Bertie Vidgen, Dirk Hovy, and Janet Pierrehumbert. Two contrasting data annotation paradigms for subjective NLP tasks. In Marine Carpuat, Marie-Catherine de Marneffe, and Ivan Vladimir Meza Ruiz (eds.), *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 175–190, Seattle, United States, July 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.naacl-main.13. URL <https://aclanthology.org/2022.naacl-main.13/>.
- Pratik S Sachdeva, Renata Barreto, Claudia von Vacano, and Chris J Kennedy. Assessing annotator identity sensitivity via item response theory: A case study in a hate speech corpus. In *Proceedings of the 2022 ACM conference on fairness, accountability, and transparency*, pp. 1585–1603, 2022.
- Maarten Sap, Dallas Card, Saadia Gabriel, Yejin Choi, and Noah A. Smith. The risk of racial bias in hate speech detection. In Anna Korhonen, David Traum, and Lluís Màrquez (eds.), *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 1668–1678, Florence, Italy, July 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1163. URL <https://aclanthology.org/P19-1163/>.
- Maarten Sap, Swabha Swayamdipta, Laura Vianna, Xuhui Zhou, Yejin Choi, and Noah A. Smith. Annotators with attitudes: How annotator beliefs and identities bias toxic language detection. In

- Marine Carpuat, Marie-Catherine de Marneffe, and Ivan Vladimir Meza Ruiz (eds.), *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 5884–5906, Seattle, United States, July 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.naacl-main.431. URL <https://aclanthology.org/2022.naacl-main.431/>.
- Archit Sharma, Sedrick Keh, Eric Mitchell, Chelsea Finn, Kushal Arora, and Thomas Kollar. A critical evaluation of ai feedback for aligning large language models. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 29166–29190. Curran Associates, Inc., 2024. doi: 10.52202/079017-0919. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/33870b3e099880cd8e705cd07173ac27-Paper-Conference.pdf.
- Ole Tange. Gnu parallel 20250822, August 2025. URL <https://doi.org/10.5281/zenodo.16944306>. GNU Parallel is a general parallelizer to run multiple serial command line programs in parallel without changing them.
- Alexandra Uma, Dina Almanea, and Massimo Poesio. Scaling and disagreements: Bias, noise, and ambiguity. *Frontiers in Artificial Intelligence*, Volume 5 - 2022, 2022. ISSN 2624-8212. doi: 10.3389/frai.2022.818451. URL <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2022.818451>.
- United Nations. Targets of hate, 9 2025. URL <https://www.un.org/en/hate-speech/impact-and-prevention/targets-of-hate>. Accessed: 2025-09-22.
- Mariken A. C. G. van der Velden, Felicia Loecherbach, Wouter van Atteveldt, Antske Fokkens, Myrthe Reuver, and Kasper Welbers. Whose truth is it anyway? an experiment on annotation bias in times of factual opinion polarization. *Communication Methods and Measures*, 19(4):332–349, 2025. doi: 10.1080/19312458.2025.2562034. URL <https://doi.org/10.1080/19312458.2025.2562034>.
- Nikolas Vitsakis, Amit Parekh, Tanvi Dinkar, Gavin Abercrombie, Ioannis Konstas, and Verena Rieser. iLab at SemEval-2023 task 11 le-wi-di: Modelling disagreement or modelling perspectives? In Atul Kr. Ojha, A. Seza Doğruöz, Giovanni Da San Martino, Harish Tayyar Madabushi, Ritesh Kumar, and Elisa Sartori (eds.), *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, pp. 1660–1669, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.semeval-1.231. URL <https://aclanthology.org/2023.semeval-1.231/>.
- Nikolas Vitsakis, Amit Parekh, and Ioannis Konstas. Voices in a crowd: Searching for clusters of unique perspectives. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 12517–12539, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.696. URL <https://aclanthology.org/2024.emnlp-main.696/>.
- Franziska Weeber, Vera Neplenbroek, Jan Batzner, and Sebastian Padó. One persona, Many cues, Different results: How sociodemographic cues impact LLM personalization. In Maria Liakata, Viviane P. Moreira, Jiajun Zhang, and David Jurgens (eds.), *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 44892–44921, San Diego, California, United States, July 2026. Association for Computational Linguistics. ISBN 979-8-89176-390-6. doi: 10.18653/v1/2026.acl-long.2079. URL <https://aclanthology.org/2026.acl-long.2079/>.
- Michael Wojatzki, Tobias Horsmann, Darina Gold, and Torsten Zesch. Do women perceive hate differently: Examining the relationship between hate speech, gender, and agreement judgments. 2018.
- Ka Wong, Praveen Paritosh, and Lora Aroyo. Cross-replication reliability - an empirical approach to interpreting inter-rater reliability. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli (eds.), *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1:*

Long Papers), pp. 7053–7065, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.548. URL <https://aclanthology.org/2021.acl-long.548/>.

An Yang et al. Qwen2.5 technical report, 2025. URL <https://arxiv.org/abs/2412.15115>.

A ACRONYMS

IAA Inter-Annotator Agreement
InPol Inherent Polarization
nDFU normalized Distance From Unimodality
NLP Natural Language Processing
GPLv3 GNU General Public Licence v3
AA African American
PC Personal Characteristic
HS Hate Speech
SD Standard Deviation

B EXPLORATORY ANALYSIS

The number of annotators per item for all datasets presented in §3 can be found in 6, and descriptive statistics can be found in Table 5. The DICES-990 dataset has exactly 123 annotations per item, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few items that vastly exceed the promised 5 number of annotators. We do not include these items, since they are very likely a data error. Tables 6,7,8 show the annotation counts for each PC group in each dataset.

C STATISTICAL ESTIMATION

Since annotation histograms may be noisy, we additionally assess whether the observed $apunim$ value could arise by chance. To assess statistical significance, we construct a null distribution of $apunim$ values from the randomized partitions and compare the observed $apunim$ against this distribution using a one-sample Student- t test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$\begin{aligned} H_0 : \bar{apunim}_{rand}(d, g) &= apunim(d, g) \\ H_a : \bar{apunim}_{rand}(d, g) &\neq apunim(d, g) \end{aligned} \quad (8)$$

where $\bar{apunim}_{rand}(d, g)$ denotes the average $apunim$ values obtained from t randomized annotator partitions.

We account for the presence of multiple subgroups in each PC dimension using a multiple-hypothesis correction. Since our test is parametric (App. C), it assumes that there exist (1) enough

Dataset	count	mean	SD	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8	1.17	69	72	73	74	76
Sap	585	5.6	.78	3	6	6	6	12
Kumar	2998	5.02	.34	5	5	5	5	10

Table 5: Number of annotations per item in each of the four datasets used in this study.

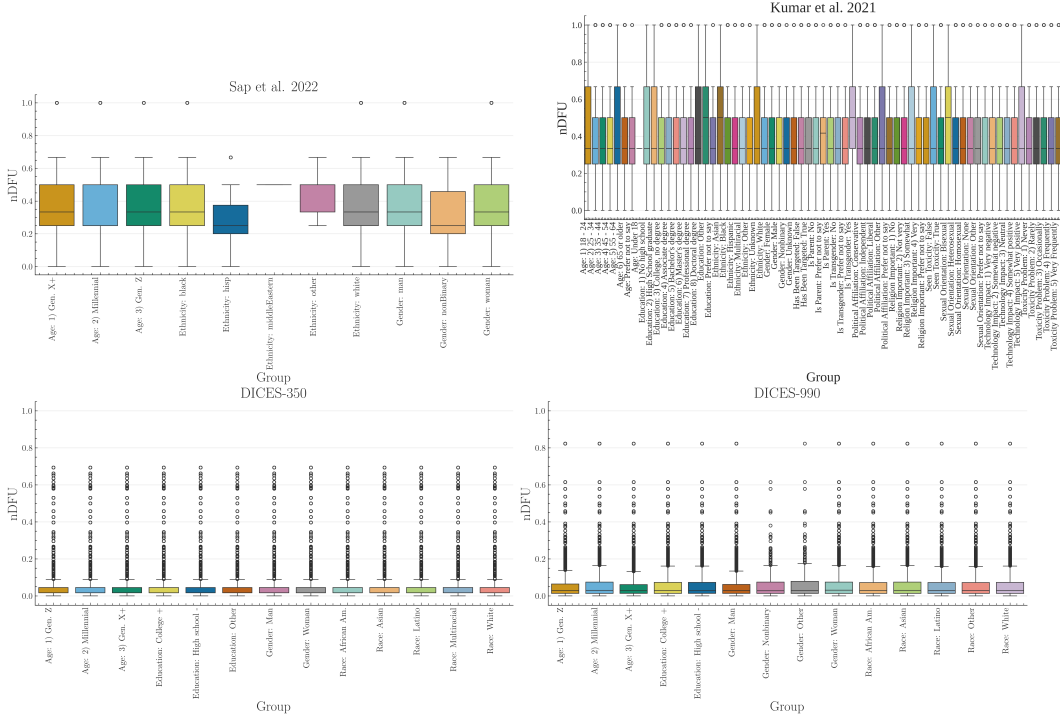


Figure 5: Boxplots of dataset polarization for the human datasets considered in this paper for all groups in each examined PC dimension.

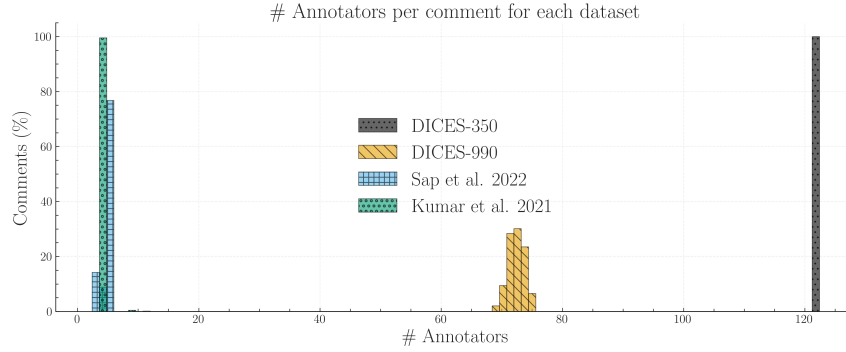


Figure 6: We calculate the ratio of items for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few items in the Kumar dataset exceeding this threshold (see Table 5), which is likely caused by a data error.

data-points to reliably run the means-test, (2) enough annotations per PC group to estimate nDFUs in Equations 4-6, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3). We discuss condition (2) in App. J. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific dimension. Since we are simultaneously considering multiple

Category	Subgroup	Support	Category	Subgroup	Support
Gender	Woman	36050	Gender	Woman	21700
	Man	35392		Man	21350
	Nonbinary	331	Race	White	10500
	Other	330		African Am.	10150
Race	Other	26394		Asian	9100
	Asian	19885		Latino	7700
	White	12322		Multiracial	5600
	Latino	7170	Age	1) Gen. Z	19600
	African Am.	6332		2) Millennial	12600
Age	Gen. Z	14899		3) Gen. X+	10850
	Millennial	41542	Education	College +	26250
	Gen. X+	15662		High school -	14350
Education	College +	64054		Other	2450
	High school -	8049			

Table 6: Subgroup Counts for DICES-990 (left) and DICES-350 (right). “Support” refers to annotation count.

Category	Subgroup	Support
Age	Gen. Z	252
	Millennial	2259
	Gen. X+	786
Ethnicity	white	2168
	black	1091
	hisp	32
	other	5
	middleEastern	1
Gender	man	1859
	woman	1412
	nonBinary	26

Table 7: Subgroup Counts for Sap et al. 2022. “Support” refers to annotation count.

hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method (Holm, 1979). The test is parameterized by the confidence level of our test, which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses (Chen et al., 2017)).

D SOFTWARE

D.1 INSTALLATION AND USAGE

We release the `apunim` package, which is a Python library that implements both the `nDFU` and `apunim` calculations, as well as its statistical test. The software is available on PyPi. We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (<https://anonymous.4open.science/r/apunim-page>).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project’s repository (<https://anonymous.4open.science/r/aposteriori-unimodality-E8C1>). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy`, `scipy`, `statsmodels`).

Category	Subgroup	Support	Category	Subgroup	Support
Gender	Female	8011	Is Trans	No	14549
	Male	6832		Yes	363
	Unknown	124		N/A	148
	Nonbinary	93	Politics	Liberal	5997
Ethnicity	White	10699		Independent	4143
	Black	1940		Conservative	3981
	Asian	862		N/A	609
	Multiracial	664		Other	330
	Hispanic	418	Is Parent	Yes	7889
	Other	247		No	7015
	Unknown	230		N/A	156
Age	Under 18	3	Tech Impact	Very negative	149
	18 - 24	1711		Somewhat negative	1411
	25 - 34	5969		Neutral	1905
	35 - 44	3785		Somewhat positive	7690
	45 - 54	1977		Very positive	3905
	55 - 64	1101	Tox Problem	Never	869
	65 or older	471		Rarely	2485
	N/A	43		Occasionally	5307
Education	No high school	71		Frequently	4441
	High School	1353		Very Frequently	1958
	College	3090	Rel Important	No	4735
	Associate's	1676		Not very	1808
	Bachelor's	6071		Somewhat	3548
	Master's	2261		Very	4741
	Professional	236		N/A	228
	Doctoral	198	Seen Toxicity	True	11505
	N/A	71		False	3555
	Other	32	Targeted	False	10741
Sexual Orient.	Heterosexual	12391		True	4319
	Bisexual	1665			
	Homosexual	515			
	N/A	302			
	Other	131			

Table 8: Subgroup Counts for the Kumar dataset. “Support” refers to annotation count.

D.2 IMPLEMENTATION DETAILS

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple PC dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{r}_i$, which are originally calculated to obtain the apriori polarization values (see Eq. 4), for the p-value calculations (App. C). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{r}_i$ in Eq. 4), and the number of apunim values used in the p-value calculation have to be the same.

We use version 1.0.6 of the apunim library for our experiments. We use 100 random iterations for the apriori value calculation (Eq. 4) and the p-value estimation (App. C), and set a confidence level of $\alpha = 0.05$. The relative library setup is provided as a conda environment file to ensure reproducibility.

While the library efficiently handles large numbers of annotators through vectorized operations, performance degrades significantly when processing a high volume of items, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §3 complete within a minute, the Kumar dataset necessitates approximately one hour of computation

using commercial hardware. Furthermore, the annotator variance analysis detailed in App. J required over ten hours. While this latter time requirement is anticipated due to the inherently demanding nature and non-standard application of our library, it demonstrates the existence of a potential scaling bottleneck. The same applies to a lesser extent for the subsampling experiments on the DICES datasets, taking collectively three hours of computation.

We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each item, for each PC dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each PC subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` package (Tange, 2025) for our own experiments). We also note that the Kumar dataset also requires substantial RAM resources, estimated around 42–46GB of dedicated RAM should the whole dataset be considered (this is not a requirement for loading the sample used in this study). The CPU-bound experiments require approximately 3 hours to run, assuming a computer capable of efficient multiprocessing.

Finally, we note that in Fig. 11, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of items in DICES-990 were annotated by more than 73 annotators—see App. B).

E APUNIM DESIGN DECISIONS

Defining Apriori Polarization In §4.1, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c | g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 2, left, we would end up subtracting the $nDFU$ s of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

Filtering Ineligible items Because *apunim* is fundamentally designed to quantify polarization, we exclude items that exhibit no measurable polarization ($nDFU(c) = 0$). Furthermore, we also exclude items annotated exclusively by a single group (i.e., items where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of $nDFU$ (see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only 5 – 6 annotations per item. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.

F WHAT IS INPOL?

Why Does it Exist? Unlike disagreement, InPol does not provide any information, but rather represents noise or uncertainty translated in terms of polarization. In order to minimize it, we need to find its cause. Since the DICES datasets are the only ones without *any* items with InPol leads us to two hypotheses:

Hypothesis 1 (Low annotator count). A low number of annotators per item may provide our model with insufficient information.

Hypothesis 2 (Inaccessible groups). InPol may arise from the presence of partially represented groups within the annotator pool which may go undetected if they are represented by only one or two annotators, as $nDFU$ is undefined for these groups.

Both of these explanations are plausible and reflect known limitations of most NLP datasets. Unfortunately, we can not confirm any of these hypotheses

The fact that the DICES datasets (featuring 70+ annotators per item) exhibit zero InPol in all items may initially seem to validate either of the above hypotheses. However, even when repeating the experiment using random annotator subsets (ranging from 5 to 50 annotators) and taking the maximum across ten random samples for each subset size, *we still observed zero InPol* (App. J). This

Content Warning: Slurs, offensive, toxic and hateful speech.	
Comment	InPol
The minks mentioned in the headline are indeed "rising" from where they were based, but that is due to them being buried in shallow mass graves (Africa) and filling with gas that causes them to rise out. Rest assured, they're still very much dead, thankfully!.	1
There should be such thing as a b***ch resistance for us women	1
There's a big difference between being a b***ch and being a woman	1
i just watched a video of @joisnotonfire being an annoying h***e on skype i'm laughing	1
n***as love to commit crime	1
I swear to God above I love ya and when ya be with this brotha, I swear I love you my bro as long as you not a n***a	0.5
I will become a cold ass b***ch because of my treatment	0.5
I would happily lay out a n***a	0.5
Shoot yourself in the head you hypersensitive fuckin b***ch.	0.5
not all women are b***ches, but all b***ches are women	0.5
"you ugly b***ch!!!"- that seems to be my real name these days"	0
"I am happy to report that drowning does kill this demonic entity. Oh holy are the waters of Chernarus!"" Omg, DED ;)"	0
"Let them praise the name of the LORD: for his name alone is excellent; his glory is above the earth and heaven."" Psalms 148:13 #Bible"	0
someone plz get this n***er some help. He's playing up like mad	0
stop talking you fucking n***er !	0

Table 9: Sample of comments from the Sap dataset and their InPol. Note that unambiguously toxic comments can be assigned with zero InPol, while more nuanced cases tend to have higher values.

finding indicates that the DICES datasets may be structurally different. This explanation is plausible given that the annotation targets are LLM texts, which may be relatively unsubtle compared to human items, as supported by low overall polarization (Fig. 5; App B). Although a definitive answer to this question eludes us due to the lack of other datasets with high annotator counts per item, the indication that the DICES datasets may not feature similar polarization patterns is noteworthy.

How Does it Look Like? Table 9 shows a random sample of items taken from the Sap dataset and their respective InPol values (§4.2). Note that explicitly toxic or racist items tend to have InPol values of zero, alongside clearly unproblematic items. Inversely, more ambiguous and less comprehensible items tend to have larger InPol values. This pattern may suggest that InPol may be tied to the overall ambiguity of the statements, especially in cases where items may either be interpreted as very toxic, or non-toxic, which agrees with the overall goal of nDFU (§2). We include the InPol values for each individual item for all four datasets in the supplementary material.

G SYNTHETIC EXPERIMENTS

This appendix details the synthetic experiments used to illustrate the functionality and behavior of nDFU. In all simulations, the bin size is set to $N_{bins} = 10$.

Disagreement vs. Polarization To explore the relationship between annotation polarization and disagreement in Fig. 1, we constructed a 2×2 matrix of synthetic datasets, each drawing from a pool of $N_{annotators} = 100$. Polarization is simulated by defining a minority group, which constitutes 20% of the total annotators ($n_{special} = 20$). These minority annotators are assigned a targeted mean shift ($mean_shift = 1$) from the base distribution. The base variance for these distributions is set at variance = 0.3. Different types of distributions (unimodal, bimodal, multimodal, uniform) are achieved by controlling the base_means and applying variance multipliers (e.g., variance $\times 3$ for multimodal).

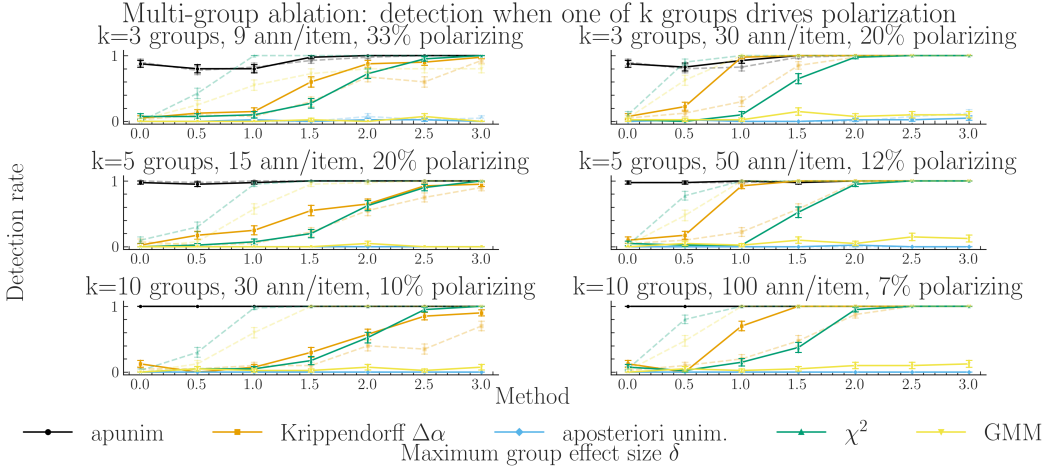


Figure 7: Polarization group attribution on synthetic distributions. Bars indicate standard error. The setup is identical to the one used for Fig. 4, although in these experiments we consider multiple k annotator groups for each item. Dashed lines show the results of a simplified polarization setup, where polarization always has the same direction.

Single-Item Polarization We simulate the distribution of toxicity annotations in Fig. 2, left, across two demographic groups (Men and Women). The distribution of Men’s scores is modeled using a truncated normal distribution with loc = 2 and scale = 1.3, while Women’s scores are modeled with loc = 8 and scale = 1.3. The combined dataset, representing all annotations, is used to demonstrate the impact of population weighting on the overall nDFU score. The sample size for each group in this visualization is $N_{annotators} = 50$.

Discussion-Level Polarization We examine nDFU scores for three specific textual examples presented in Fig. 3. For the individual items, $N_{annotators} = 200$ for each group. In the first case (misogynist item), Men’s scores are modeled with loc = 6, while Women’s scores are modeled with loc = 2. In the second case (misandrist item), the roles are reversed. The third visualization combines distributions for men and women derived from the respective single-item distributions.

H EXPERIMENT ABLATIONS

Why Do DICES not Exhibit InPol? In order to investigate whether the numerous annotators per items can explain the lack of InPol in the DICES datasets (§4) we gradually subsample the annotators for each item randomly with replacement, starting from five annotators per item, and scaling up to the full complement of annotators in the dataset with a step of two. For each such step, we choose ten annotator samples and compute the maximum InPol across all runs. Even with annotator counts similar to the Kumar and DICES datasets, InPol remains zero, suggesting that the absence of InPol can not be attributed to annotator number alone—rather, it is plausibly a pattern attributable to the DICES datasets themselves. This kind of analysis may be used in the exploratory phase of dataset selection in order to filter out datasets without enough group-specific subjectivity.

Does Apunim Work for More than Two Groups? We repeat the experiment comparing apunim to prior methodologies shown in Fig. 4 but generalized across multiple groups. As shown in Fig. 7, apunim stays competitive even in the presence of multiple annotator groups. Note the increased efficiency of many methods when polarization is unidirectional. This pattern indicates that the challenges of bidirectional polarization severely limiting prior approaches may be generalizable to more than two groups.

Table 10: Comparison of Apunim values across different sample sizes (1k, 3k, 10k, 30k) for the Kumar dataset.

		1000	3000	10000	30000
	PC				
Age	1) 18 - 24	0.014	-0.017	-0.037	-0.016
	2) 25 - 34	-0.003	-0.000	0.000	0.000
	3) 35 - 44	0.000	-0.011	-0.001	-0.001
	4) 45 - 54	0.022	0.033	0.013	0.003
	5) 55 - 64	0.000	0.000	-0.057	0.001
	6) 65 or older	—	—	—	0.948
Education	2) High School graduate	0.000	0.000	-0.005	0.014
	3) College, no degree	0.010	0.014	0.010	0.006
	4) Associate degree	0.000	0.000	0.005	0.009
	5) Bachelor’s degree	0.001	-0.000	-0.001	-0.000
	6) Master’s degree	0.000	-0.006	0.004	-0.000
Ethnicity	Asian	-0.000	0.000	0.011	0.007
	Black	0.000	-0.024	-0.004	-0.012
	Hispanic	—	—	0.000	-0.099
	Multiracial	0.000	0.000	0.000	0.000
	Other	—	—	—	-0.306
	White	0.000	-0.000	-0.000	0.000
Gender	Female	0.001	0.000	0.002	0.001
	Male	0.001	-0.004	-0.002	-0.001
Has Been Targeted	False	0.001	0.000	-0.000	0.000
	True	-0.009	-0.015	-0.003	-0.002
Is Parent	No	-0.002	0.002	0.001	0.001
	Prefer not to say	—	—	1.000	0.000
	Yes	-0.000	0.000	-0.000	0.000
Is Transgender	No	0.000	0.000	0.000	0.000
Political Affiliation	Conservative	-0.013	0.003	-0.003	-0.002
	Independent	0.000	-0.001	0.001	0.002
	Liberal	-0.001	0.006	0.001	0.001
	Other	—	0.000	0.000	0.000
	Prefer not to say	—	—	0.000	0.022
Religion Important	1) No	0.002	0.004	0.002	0.001
	2) Not very	0.000	-0.006	0.010	0.006
	3) Somewhat	0.000	0.002	0.003	-0.002
	4) Very	0.000	-0.003	-0.003	-0.002
Seen Toxicity	False	-0.007	0.002	-0.001	-0.001
	True	-0.001	0.000	0.001	0.000
Sexual Orientation	Bisexual	-0.068	-0.029	-0.040	-0.013
	Heterosexual	-0.001	-0.000	-0.000	0.000
	Homosexual	—	—	0.000	0.053
	Prefer not to say	—	—	—	0.370
Technology Impact	2) Somewhat negative	0.000	0.000	0.000	0.003
	3) Neutral	0.124	0.037	-0.014	0.000
	4) Somewhat positive	0.004	0.001	-0.000	0.001
	5) Very positive	-0.015	-0.004	-0.004	-0.003
Toxicity Problem	1) Never	0.000	0.000	-0.037	-0.081
	2) Rarely	0.000	0.000	0.003	-0.001
	3) Occasionally	0.000	-0.004	-0.003	-0.001
	4) Frequently	0.000	0.000	-0.001	-0.000
	5) Very Frequently	0.000	-0.002	0.030	0.015

How Does Dataset Size Influence Apunim? Since our approach was designed for the small-scale datasets common in NLP, we were cautious when testing it on the massive Kumar dataset. Because

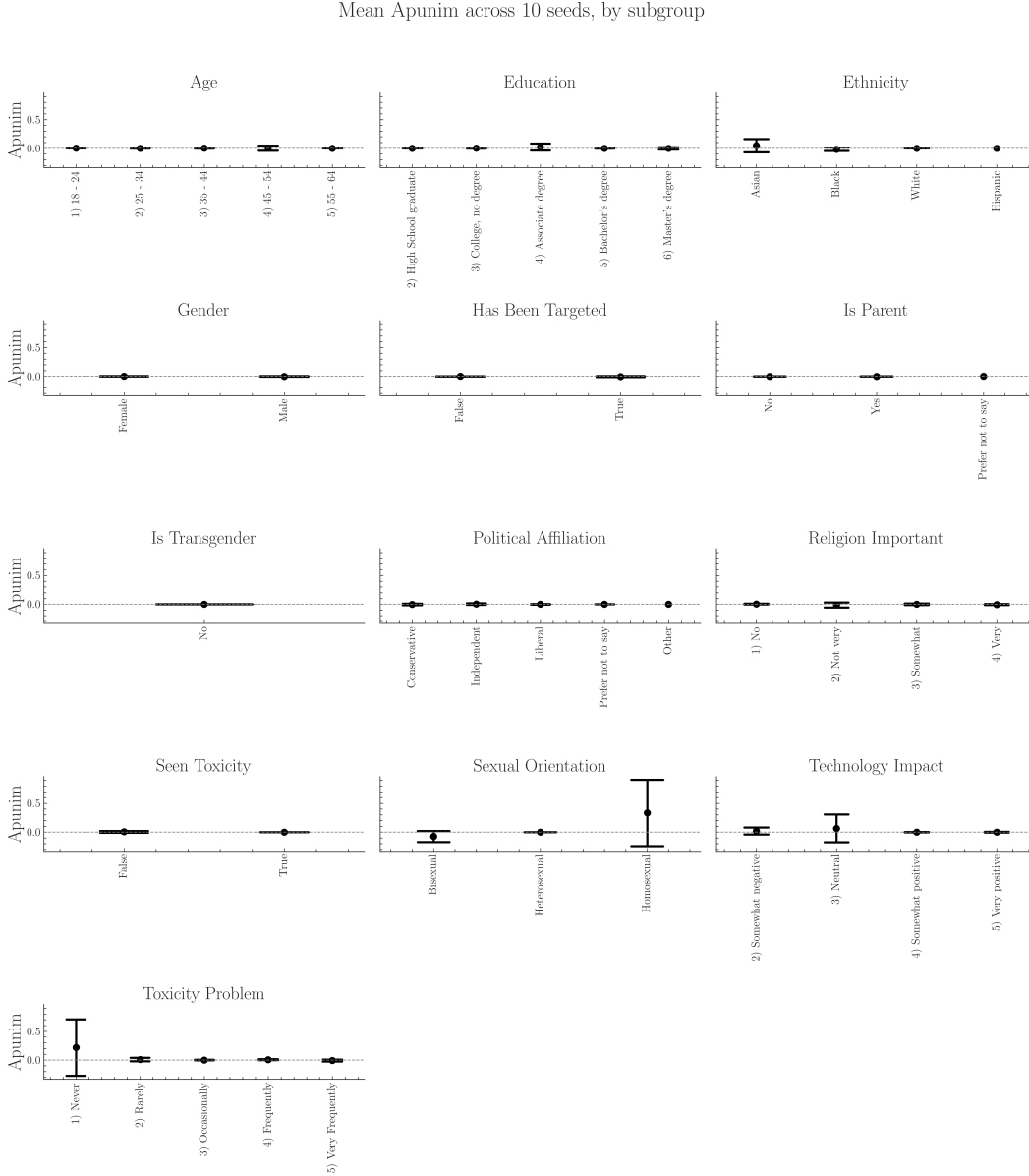


Figure 8: Apunim values on the Kumar dataset ($n_{\text{samples}} = 1000$) across ten runs with different subsets. Black bars indicate two SDs. PC dimensions with many groups are more volatile, although all dimensions broadly keep the same apunim sign across subsets.

of the data volume, we analyzed various smaller samples (1,000, 3,000, 10,000, and 30,000 items), shown in Table 10. Sample size does not seem to have a consistent effect on apunim values, other than large splits uncovering significant polarization effects in sparsely populated groups (e.g., 65+ yo. in Age, “Prefer Not to Say” in *Sexual Orientation*, “Never”, “Very Frequently” in *Toxicity Problem*). The apunim direction stays broadly the same. We also test whether the results in Table 2 are an artifact of the dataset split, because of the small sample size used in the study. We repeat the apunim experiment with ten different subsets of the same dataset and record their apunim values, shown in Fig. 8. We find that most apunim values are stable, save for “Never” in *Toxicity Problem*, “Heterosexual” in *Sexual Orientation* and “Neutral” in *Technology Impact*.

How Does Sample Selection Influence Apunim? Astute readers may notice that some variability does exist across differently sized samples for the same examined groups, where for example the

sign of the apunim value may switch. We note that the statistically significant values (bold rows) change signs only when the apunim value approaches zero in the largest samples. Thus, the results are not inverted, but rather pushed towards zero. In order to verify that this variability is caused by the sample size, and not by different samples being drawn, we execute a further ablation study. Specifically, we execute the apunim study on the 3000-sized sample of the dataset, using different subsets of the Kumar dataset. The results can be found in Fig. 8, where we note that variability is not an issue, except for a couple of cases. These cases seem to represent sparse annotator groups, where missing a few items may genuinely change the value of the apunim metric. These groups however are largely not detected by apunim as significant either way (see Table 2).

I LLM ANNOTATION PROCEDURE

I.1 MODELS USED

We use only open-source models, as proprietary models such as OpenAI’s GPT-5 yield much less reproducible results (Bisbee et al., 2024). Furthermore, the use of open-source models requires much fewer resources, enabling research on this task with a less demanding budget, as well as scalable deployment in real-world settings (Rossi et al., 2024). It is also worth noting that since the datasets used are not publicly licensed, we can not risk leaking the data contained within to proprietary models. Generally, the use of local models protects the personal data of the users contained in our datasets, ensures that our research can be replicated in domains where anonymity must be maintained, and renders our work compatible with existing legislation (European Union, 2024; 2016). Table 11 shows details on the LLMs used for this study. We specifically use 4-bit quantized versions from the `unsloth` `huggingface` repository.

Family	Version	#Params
OLMo (Groeneveld et al., 2024)	2 (0325)	32B
Qwen (Yang et al., 2025)	2.5	32B
LLaMa (Grattafiori et al., 2024)	3.3	70B
OLMo (Olmo, 2026)	3	7B
Qwen (Yang et al., 2025)	2.5	7B
LLaMa (Grattafiori et al., 2024)	3.1	8B

Table 11: LLMs used in this study.

I.2 EXPERIMENTAL SETUP

Since each of the datasets in §3 was annotated using different annotation guidelines and targets we use different instruction prompts for each dataset. The “default” annotation prompt is directly extracted from the studies in the case of Sap and Kumar, and inferred from the instruction guidelines used in the DICES datasets with minor modifications. We use standard persona prompting procedures sampled randomly from the available PC dimensions in each respective dataset. Six LLM annotators are selected for each item and model.

The prompts used in the ablation study were provided by a GPT-5 and a local Gemma-4 (quantized) model by providing the original prompt and instructing it to produce four variants. The tested prompts use different methods to restrict the model’s behavior. The initial prompts (“*Persona*”) require the model to adopt a persona’s viewpoint to judge the comment. Other prompts (“*direct*”) treat the persona’s characteristics as an absolute, non-negotiable rule that overrides standard moderation guidelines. Another approach (“*single*”) compels the model to perform a persona-based check before submitting its final rating. The most aggressive prompts (“*stereotype*”) force the model into extreme decision spaces. The full main and ablation annotation prompts can be found in App. L.

We use a single Quadro RTX 6000 GPU for tasks requiring small (7-8B) models and two such GPUs for the rest. The annotation task, including ablations, requires approximately 300 hours of computation to complete. We use the `transformers` library to load and manage the LLMs. We release the relevant code in the supplementary material. Our codebase is platform-independent except for three high-level bash scripts and a full environment file is included for reproduction.

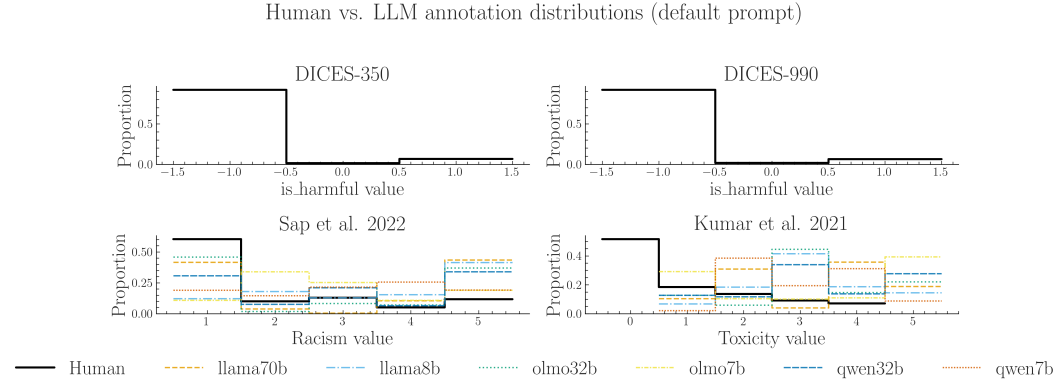


Figure 9: Annotation frequencies for humans and each LLM in the primary annotation task using the default prompt.

Table 12: Consistency (Krippendorff’s α , ordinal) of each model with itself across repeated runs of the same (default) prompt.

Dataset	Model	Runs	Items	Krippendorff’s alpha
Sap et al. 2022	llama70b	5	180	1.0000
Sap et al. 2022	llama8b	5	180	1.0000
Sap et al. 2022	olmo32b	5	178	1.0000
Sap et al. 2022	olmo7b	5	172	1.0000
Sap et al. 2022	qwen32b	5	180	1.0000
Sap et al. 2022	qwen7b	5	180	1.0000
Kumar et al. 2021	llama70b	5	600	1.0000
Kumar et al. 2021	llama8b	5	600	1.0000
Kumar et al. 2021	olmo32b	5	600	1.0000
Kumar et al. 2021	olmo7b	5	567	1.0000
Kumar et al. 2021	qwen32b	5	600	1.0000
Kumar et al. 2021	qwen7b	5	589	1.0000

I.3 EXPLORATORY ANALYSIS

Fig. 9 shows the different annotation frequencies of LLMs and human annotators in the same dataset split. Across all four datasets, we observe a consistent mismatch between human and LLM annotation distributions. For the DICES-350 and DICES-990 harmfulness tasks, human annotations are heavily concentrated at the lowest harmfulness value, whereas most LLMs assign substantially more probability to moderate and high harmfulness values. A similar pattern appears in the Sap et al. (2022) racism dataset: human judgments are concentrated at low racism scores, while LLMs produce broader distributions with considerably more mass at higher scores, particularly values 4–5. The (Kumar et al., 2021) toxicity results show the same general trend, with humans assigning most probability to zero toxicity while LLMs distribute their predictions across higher toxicity levels, often peaking around values 2–4. Importantly, the magnitude of this discrepancy varies across models; some LLMs are closer to the human distribution, while others show substantially stronger shifts toward higher scores.

We also investigate whether LLM annotations include any InPol. Table 3 shows that InPol is significantly smaller than in human annotations. Adversarial prompts seem to consistently increase it, although it does not approach human levels. It is also notable that as with LLM apunim values (next section), InPol seems to be primarily model-dependent.

I.4 ANNOTATION ABLATIONS

Table 13: Consistency (Krippendorff’s α , ordinal) of each model with itself across the three paraphrased prompt variants.

Dataset	Model	Variants	Items	Krippendorff’s alpha
Sap et al. 2022	llama70b	3	180	0.8829
Sap et al. 2022	llama8b	3	180	0.6717
Sap et al. 2022	olmo32b	3	180	0.9086
Sap et al. 2022	olmo7b	3	180	0.7090
Sap et al. 2022	qwen32b	3	180	0.9199
Sap et al. 2022	qwen7b	3	180	0.7993
Kumar et al. 2021	llama70b	3	600	0.8406
Kumar et al. 2021	llama8b	3	600	0.3988
Kumar et al. 2021	olmo32b	3	600	0.7827
Kumar et al. 2021	olmo7b	3	593	0.3897
Kumar et al. 2021	qwen32b	3	600	0.9457
Kumar et al. 2021	qwen7b	3	600	0.5578

Table 14: Consistency (Krippendorff’s α , ordinal) between LLMs given the same (default) prompt, per dataset.

Dataset	Prompt	Models	Items	Krippendorff’s alpha
Sap et al. 2022	default	6	3510	0.6556
Kumar et al. 2021	default	6	7741	0.4519

We examine the following conditions for selecting models and datasets for our LLM annotation experiment. All ablation experiments use a sampled 20% fraction of the full LLM annotation dataset used in the main text.

The LLMs should be consistent given the same prompt While we have fixed all relevant seeds, it is possible that generation deviates across repeated model prompting given the same input (Bisbee et al., 2024). We check this condition by providing the model with the main annotation prompt (see App. L) five times and find that generation is indeed deterministic under our setup for all models (Table 12).

The LLMs should be prompt insensitive Small changes in the prompt have been shown to drastically affect LLM output in some tasks depending on the experimental setup (Razavi et al., 2025). In order to verify whether any models are unreliable under our setup, we create two alternative formulations of each prompt without a change in substance using a GPT-5 model (free version), whose outputs we manually verified (see App. L). Table 13 shows that while most LLMs are relatively robust to changes in formulation, the smaller ones (OLMo-7B and LLaMa-8B) are not. They are thus excluded from the annotation study.

The annotation guidelines should be unambiguous The studies that provided the datasets used in our work (see §3) do not include LLM annotation, nor equivalent LLM prompts, which meant we had to generate them from the human annotation guidelines ourselves. This was not a problem in the case of the Sap and Kumar datasets, since the annotation guidelines were short, simple and required minimal edits to be transformed into LLM annotation guidelines. However, the DICES datasets use poorly documented guidelines and the target label we used is an aggregation of multiple annotation tasks. For these reasons the prompt creation is non-trivial and prone to errors. Indeed, Table 14 shows that inter-model agreement is extremely low for the DICES datasets, even when we remove the sensitive models (Table 15). Given that the DICES datasets are (1) problematic w.r.t. inherent polarization (see App. G), (2) conditioned on several dozen annotations per item (3) difficult to find reliable annotation prompts for LLMs, we decided to exclude them from the annotation study.

Table 15: Consistency (Krippendorff’s α , ordinal) between LLMs given the same (default) prompt, per dataset, excluding the following models: llama8b, olmo7b.

Dataset	Prompt	Models	Items	Krippendorff’s alpha
Sap et al. 2022	default	4	3510	0.7594
Kumar et al. 2021	default	4	6000	0.5999

I.5 LLM ANNOTATION RESULTS

Overall Polarization Fig. 10 shows the overall polarization patterns when using the different adversarial instructions. The Qwen-23B and LLaMa-70B models consistently produce no polarization, while the persona-focused prompt seems to elicit some varied polarization, even though it does not translate to higher apunim values. This pattern would indicate that the persona-focused prompt likely adds more overall variance to the persona-conditioned annotations, even though that variance is not consistent.

Apunim Analysis Tables 16 and 17 show the apunim results for the LLM annotation task on the Kumar and Sap datasets respectively. Apunim values are largely model-dependent, but no other concrete pattern can be discerned.

Table 16: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
Age	1) 18 - 24	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	2) 25 - 34	llama70b	—	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	3) 35 - 44	llama70b	0.000	0.000	0.000	0.000	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	4) 45 - 54	llama70b	0.000	0.000	0.000	-0.096	0.000
		olmo32b	-0.005	0.017	-0.021	0.018	-0.048
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	-0.007	0.000	0.000	-0.043	0.000
	5) 55 - 64	llama70b	0.000	0.000	—	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	-0.000	0.000	0.000
	6) 65 or older	llama70b	0.000	0.000	0.000	0.135	0.000
		olmo32b	-0.029	0.002	-0.023	0.006	-0.033
		qwen32b	0.166	—	0.000	0.000	0.000
		qwen7b	0.031	0.031	-0.012	-0.001	0.000
	Prefer not to say	llama70b	0.000	0.000	0.000	-0.130	0.000
		olmo32b	-0.016	-0.019	-0.009	-0.043	-0.003
		qwen32b	-0.222	0.000	0.000	0.000	0.000
		qwen7b	-0.033	-0.028	-0.015	0.025	0.000
	Under 18	llama70b	0.000	—	0.000	0.000	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	—
		qwen7b	0.000	0.000	0.000	0.000	0.000
Education	1) No high school	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	—
		qwen7b	0.000	0.000	0.000	0.000	0.000
	2) High School graduate	llama70b	0.000	0.000	—	0.000	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	—	—	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	3) College, no degree	llama70b	0.000	—	—	—	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	-0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000

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Table 16: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
	4) Associate degree	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	—	0.000	0.000	—	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	5) Bachelor's degree	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	—	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	6) Master's degree	llama70b	—	0.000	0.000	0.000	—
		olmo32b	—	0.000	0.000	0.000	0.000
		qwen32b	—	—	0.000	0.000	—
		qwen7b	0.000	0.000	0.000	0.000	0.000
	7) Professional degree	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	8) Doctoral degree	llama70b	0.000	0.000	—	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	—
		qwen7b	0.000	0.000	0.000	0.000	0.000
	Other	llama70b	0.000	0.000	0.000	0.000	—
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	—	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	Prefer not to say	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	—	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
Ethnicity	Asian	llama70b	0.069	0.074	0.000	0.000	0.000
		olmo32b	0.000	0.012	-0.017	0.008	0.000
		qwen32b	0.000	-0.083	0.000	—	0.000
		qwen7b	0.000	-0.023	-0.012	-0.006	-0.019
	Black	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	Hispanic	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	Multiracial	llama70b	0.000	—	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	Other	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	0.000	0.000	0.000
	Unknown	llama70b	-0.190	-0.194	0.000	0.000	0.000
		olmo32b	0.000	0.017	-0.007	0.057	0.000
		qwen32b	0.000	-0.030	0.000	-0.032	0.000
		qwen7b	0.000	-0.032	-0.022	-0.006	-0.011
	White	llama70b	0.000	0.000	—	0.000	0.000
		olmo32b	0.000	0.000	0.000	0.000	0.000
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.000	0.000	-0.000	0.000	0.000
Gender	Female	llama70b	-0.010	0.015	-0.035	0.109	-0.003
		olmo32b	0.036	0.002	0.016	-0.004	-0.017
		qwen32b	-0.109	-0.046	-0.012	-0.050	0.000
		qwen7b	0.035	0.011	0.024	0.005	-0.005
	Male	llama70b	0.003	-0.017	0.052	0.005	-0.063
		olmo32b	0.017	-0.033	0.012	0.033	0.012
		qwen32b	0.017	-0.084	-0.042	0.001	0.000
		qwen7b	-0.005	0.029	0.006	-0.014	0.024
	Nonbinary	llama70b	-0.025	-0.039	0.044	0.031	0.078
		olmo32b	-0.003	0.018	-0.013	-0.008	-0.003
		qwen32b	0.000	0.084	0.059	-0.015	0.000
		qwen7b	0.020	-0.024	-0.007	-0.003	0.015
	Unknown	llama70b	0.011	0.037	-0.011	-0.027	0.022
		olmo32b	-0.012	-0.038	-0.024	-0.028	0.008
		qwen32b	0.060	-0.026	0.012	0.019	0.000
		qwen7b	-0.044	0.003	0.047	0.029	-0.005
	Has Been Targeted	False	llama70b	0.002	-0.001	-0.024	-0.026

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Table 16: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
Is Parent	True	olmo32b	0.004	0.023	0.025	0.017	-0.003
		qwen32b	0.013	0.002	0.000	-0.032	-0.021
		qwen7b	0.008	-0.005	-0.004	0.003	0.008
		llama70b	-0.057	-0.001	0.043	0.023	-0.021
		olmo32b	-0.021	-0.021	-0.016	-0.001	0.005
		qwen32b	-0.041	0.036	-0.006	0.014	0.024
		qwen7b	-0.022	-0.015	-0.001	-0.005	-0.015
		llama70b	-0.010	0.011	-0.057	-0.034	-0.009
	No	olmo32b	-0.016	-0.017	-0.017	0.019	0.016
		qwen32b	0.045	0.024	-0.006	-0.023	-0.025
		qwen7b	-0.013	0.018	-0.024	-0.008	0.012
		llama70b	-0.021	-0.003	0.031	0.018	-0.035
		olmo32b	0.008	0.045	0.028	-0.000	-0.025
		qwen32b	-0.017	-0.047	0.111	0.036	0.035
		qwen7b	0.013	-0.018	-0.010	-0.006	-0.021
		llama70b	0.024	-0.015	0.009	0.013	0.027
	Prefer not to say	olmo32b	0.023	-0.018	-0.014	-0.013	0.005
		qwen32b	0.014	0.073	-0.037	-0.016	-0.001
		qwen7b	0.018	-0.028	-0.002	-0.011	-0.028
		llama70b	0.020	0.033	0.030	0.041	0.005
		olmo32b	0.014	0.025	-0.030	0.028	0.036
		qwen32b	0.029	0.030	0.012	-0.041	-0.024
		qwen7b	-0.019	-0.008	-0.012	0.034	-0.001
		llama70b	-0.025	-0.037	-0.047	-0.048	0.043
Is Transgender	No	olmo32b	-0.005	-0.024	0.025	-0.012	0.020
		qwen32b	0.031	-0.046	0.021	-0.020	-0.026
		qwen7b	0.039	0.002	0.017	-0.006	-0.014
		llama70b	0.016	0.040	-0.016	-0.013	-0.064
	Prefer not to say	olmo32b	0.009	0.018	-0.035	-0.005	-0.024
		qwen32b	-0.103	0.029	-0.019	0.001	0.029
		qwen7b	-0.039	0.003	0.024	-0.027	0.029
		llama70b	0.000	0.000	0.034	-0.018	-0.017
	Yes	olmo32b	-0.003	0.019	0.013	0.015	0.025
		qwen32b	-0.084	0.000	-0.020	0.000	0.000
		qwen7b	-0.025	-0.018	0.028	0.010	0.013
		llama70b	0.000	0.000	0.047	-0.031	0.000
		olmo32b	0.009	-0.018	0.019	0.019	0.001
		qwen32b	-0.054	0.000	0.000	0.000	-0.038
		qwen7b	-0.016	0.004	0.000	0.006	0.019
		llama70b	0.000	0.000	-0.075	0.026	0.000
Political Affiliation	Conservative	olmo32b	0.000	0.009	-0.021	-0.012	-0.013
		qwen32b	0.037	0.000	-0.010	0.000	-0.005
		qwen7b	-0.004	0.023	-0.015	0.006	0.034
		llama70b	0.000	0.000	0.000	-0.041	0.071
	Independent	olmo32b	-0.010	-0.005	-0.019	-0.023	0.019
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.031	-0.005	-0.003	0.009	-0.006
		llama70b	0.000	0.000	0.000	0.048	0.000
	Liberal	olmo32b	-0.006	0.023	0.009	-0.013	-0.001
		qwen32b	0.061	0.000	0.000	0.000	0.059
		qwen7b	0.015	0.017	0.019	0.000	-0.015
		llama70b	-0.007	0.017	-0.014	0.009	-0.057
	Other	olmo32b	-0.018	0.009	-0.020	0.006	-0.015
		qwen32b	-0.014	0.074	-0.067	-0.017	0.000
		qwen7b	-0.078	-0.013	-0.006	-0.007	-0.003
		llama70b	-0.021	-0.020	0.000	-0.029	0.000
Religion Important	1) No	olmo32b	-0.021	0.007	0.020	0.008	0.008
		qwen32b	0.000	0.000	0.059	0.000	0.096
		qwen7b	0.011	-0.005	0.021	0.020	0.026
		llama70b	0.000	0.000	0.000	0.000	0.000
	2) Not very	olmo32b	0.017	0.028	-0.023	-0.012	0.028
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	-0.026	-0.003	-0.006	0.017	0.000
		llama70b	0.006	0.005	-0.048	0.000	0.017
	3) Somewhat	olmo32b	0.011	0.022	0.013	0.018	0.020
		qwen32b	0.000	-0.038	0.038	0.000	0.000
		qwen7b	-0.005	-0.021	-0.025	-0.054	-0.015
		llama70b	0.015	0.049	0.136	-0.066	0.000
	4) Very	olmo32b	-0.003	-0.009	-0.004	0.002	-0.008
		qwen32b	0.092	0.000	-0.130	-0.012	-0.017
		qwen7b	0.013	0.007	0.008	0.007	-0.005
		llama70b	0.027	0.038	0.021	0.007	0.007
Seen Toxicity	False	olmo32b	-0.012	0.002	-0.013	0.006	-0.003

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Table 16: Aposteriori unimodality results for the LLM annotations of the Kumar et al. 2021 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
Sexual Orientation	True	qwen32b	0.013	-0.055	0.003	-0.069	-0.066
		qwen7b	-0.005	-0.012	0.011	-0.007	0.039
		llama70b	-0.033	-0.017	-0.002	0.005	0.085
		olmo32b	-0.011	0.029	-0.008	0.002	-0.004
		qwen32b	-0.004	0.013	0.015	0.016	0.052
	Bisexual	qwen7b	0.013	0.012	-0.011	-0.016	-0.032
		llama70b	0.000	0.000	0.000	-0.002	0.050
		olmo32b	-0.026	0.020	-0.004	-0.032	0.029
		qwen32b	0.000	0.047	0.000	0.083	-0.018
		qwen7b	-0.018	0.004	0.013	-0.007	0.027
	Heterosexual	llama70b	0.000	0.000	0.000	0.061	0.000
		olmo32b	-0.023	0.016	-0.005	0.026	0.005
		qwen32b	-0.033	-0.057	0.000	-0.021	-0.049
		qwen7b	-0.004	-0.004	0.000	0.023	0.002
	Homosexual	llama70b	0.000	0.000	0.000	-0.038	-0.026
		olmo32b	0.039	0.000	-0.009	-0.005	0.000
		qwen32b	0.000	0.000	-0.029	-0.079	0.000
		qwen7b	0.009	-0.018	-0.030	0.023	0.030
	Other	llama70b	0.000	0.000	0.000	-0.032	0.000
		olmo32b	-0.002	-0.023	0.004	0.003	0.017
		qwen32b	-0.061	0.000	0.000	0.000	0.048
		qwen7b	-0.025	-0.004	-0.002	-0.032	0.012
	Prefer not to say	llama70b	0.000	0.000	0.000	-0.028	-0.024
		olmo32b	-0.019	0.011	0.010	-0.020	0.016
		qwen32b	0.000	0.085	0.036	-0.025	0.060
		qwen7b	0.024	-0.018	-0.027	-0.009	0.042
Technology Impact	1) Very negative	llama70b	0.085	-0.010	0.000	0.000	0.000
		olmo32b	0.000	-0.013	0.000	0.000	0.016
		qwen32b	0.000	0.000	-0.061	0.000	-0.029
		qwen7b	0.000	0.000	0.000	0.008	0.000
	2) Somewhat negative	llama70b	0.000	0.000	0.000	0.000	0.000
		olmo32b	0.010	-0.026	-0.008	0.000	-0.022
		qwen32b	0.000	0.000	0.000	0.000	-0.013
		qwen7b	0.000	0.000	0.000	0.010	0.000
	3) Neutral	llama70b	0.000	0.000	-0.042	0.050	0.000
		olmo32b	-0.028	0.002	0.019	0.017	0.024
		qwen32b	-0.346	-0.017	0.000	0.000	-0.067
		qwen7b	0.000	-0.002	0.026	-0.021	-0.030
	4) Somewhat positive	llama70b	-0.017	0.000	0.059	-0.019	0.000
		olmo32b	0.007	-0.009	-0.009	-0.035	-0.001
		qwen32b	0.139	0.081	0.000	0.000	0.000
		qwen7b	0.005	0.011	-0.008	0.000	-0.023
	5) Very positive	llama70b	-0.038	-0.009	0.000	-0.006	0.000
		olmo32b	0.025	0.065	0.016	0.001	-0.024
		qwen32b	0.000	0.195	0.107	0.000	-0.014
		qwen7b	0.012	0.011	0.027	-0.022	0.015
Toxicity Problem	1) Never	llama70b	-0.049	0.001	0.062	-0.025	0.000
		olmo32b	-0.010	-0.004	-0.010	-0.014	0.029
		qwen32b	0.000	0.000	-0.007	-0.015	0.000
		qwen7b	-0.015	0.018	-0.018	0.008	-0.034
	2) Rarely	llama70b	0.000	0.000	0.078	0.027	-0.011
		olmo32b	-0.013	0.009	0.001	-0.012	-0.012
		qwen32b	0.000	-0.043	0.000	0.042	0.000
		qwen7b	-0.003	-0.004	0.017	0.022	0.000
	3) Occasionally	llama70b	-0.009	0.000	0.000	0.000	0.000
		olmo32b	0.008	0.019	0.004	0.004	-0.020
		qwen32b	0.000	0.000	0.000	0.000	0.000
		qwen7b	0.012	-0.019	0.012	-0.018	0.031
	4) Frequently	llama70b	-0.008	0.046	0.000	0.023	0.000
		olmo32b	-0.001	-0.035	-0.028	-0.005	-0.017
		qwen32b	0.000	0.000	-0.020	-0.043	0.000
		qwen7b	-0.016	-0.013	0.021	-0.017	0.010
	5) Very Frequently	llama70b	0.050	-0.046	-0.051	-0.030	-0.032
		olmo32b	-0.006	-0.022	-0.005	-0.001	0.004
		qwen32b	0.000	-0.018	-0.012	-0.031	0.000
		qwen7b	-0.010	0.010	-0.014	0.010	0.018

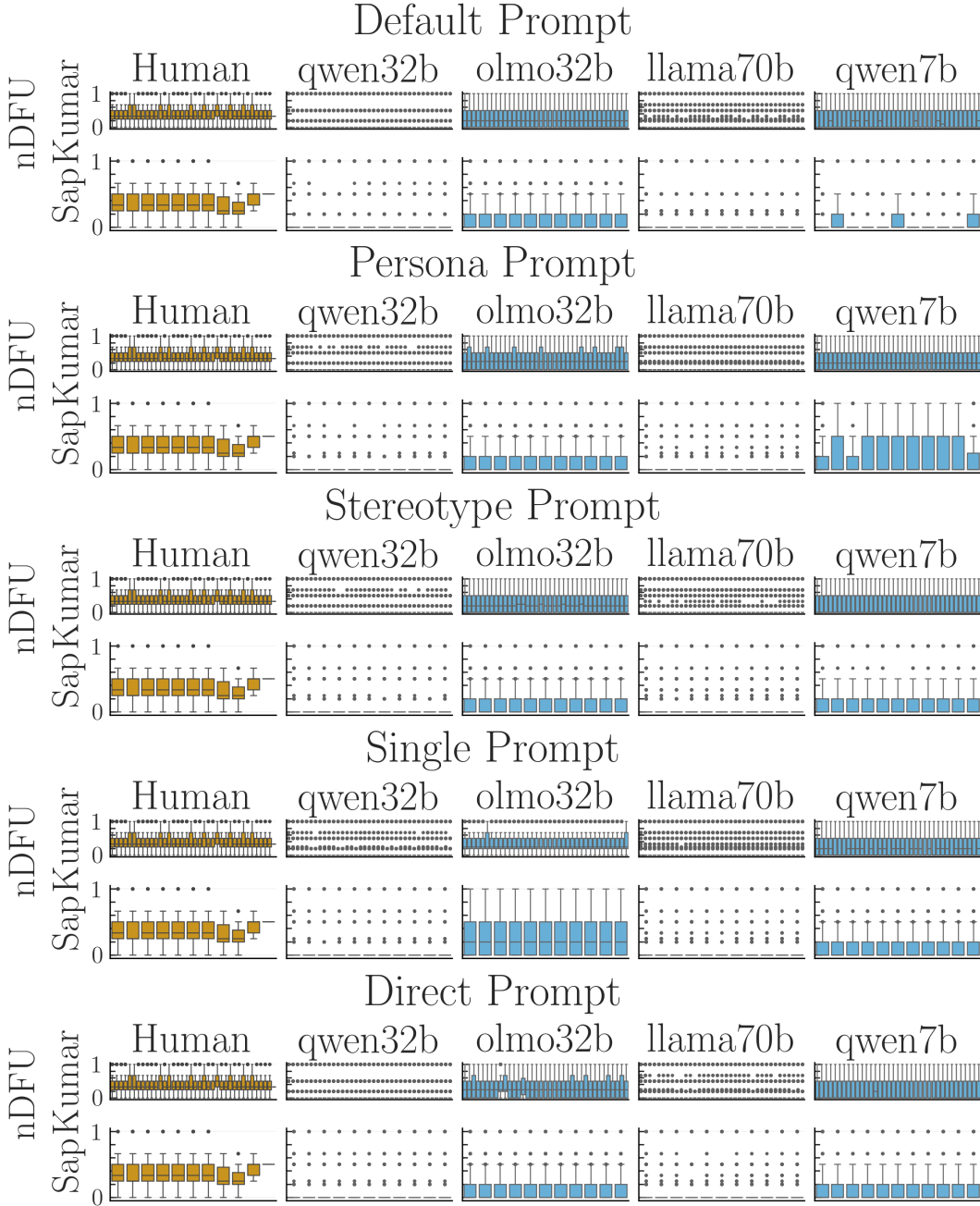


Figure 10: nDFU grouped by dataset (rows) and model family (columns) for each PC persona dimension in the primary annotation task (blue) using different instruction prompts, compared with human annotations on the same sample (yellow).

Table 17: Aposteriori unimodality results for the LLM annotations of the Sap et al. 2022 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct
Age	1) Gen. X+	llama70b	0.011	0.020	-0.078	-0.057	0.022
		olmo32b	-0.054	0.038	0.024	-0.002	0.009
		qwen32b	0.008	0.000	-0.004	-0.076	0.043

Continued on next page

Table 17: Aposteriori unimodality results for the LLM annotations of the Sap et al. 2022 dataset, across instruction prompts.

PC	Value	Model	Default	Stereotype	Persona	Single	Direct		
	2) Millennial	qwen7b	-0.033	0.021	0.013	0.007	0.036		
		llama70b	-0.069	0.016	-0.027	0.027	0.027		
		olmo32b	0.011	0.005	0.008	-0.022	0.020		
		qwen32b	-0.028	-0.041	0.016	0.076	0.068		
	3) Gen. Z	qwen7b	0.023	0.002	-0.045	0.013	-0.053		
		llama70b	0.197	0.007	0.018	-0.012	-0.041		
		olmo32b	0.001	0.023	-0.016	0.015	-0.026		
		qwen32b	0.052	-0.035	-0.017	0.070	-0.076		
		qwen7b	-0.058	-0.029	0.022	0.052	0.054		
		Ethnicity	black	llama70b	0.000	0.000	0.000	0.000	0.000
				olmo32b	-0.038	-0.048	-0.024	0.019	0.024
				qwen32b	0.021	0.000	0.000	0.000	0.000
qwen7b	0.000			-0.000	-0.000	0.000	0.035		
hisp	llama70b		0.000	0.000	-0.000	0.000	-0.000		
	olmo32b		0.131	-0.070	-0.105	0.058	-0.032		
	qwen32b		0.000	0.000	0.000	-0.000	-0.000		
	qwen7b		0.000	0.034	0.000	0.000	0.039		
middleEastern	llama70b		-0.000	-0.000	-0.000	0.000	0.000		
	olmo32b		0.181	-0.031	0.062	-0.043	0.011		
	qwen32b		0.083	0.000	0.000	0.000	0.000		
	qwen7b		-0.000	-0.058	-0.000	-0.000	-0.066		
other	llama70b	0.000	-0.000	0.000	0.000	0.000			
	olmo32b	0.000	0.000	0.000	-0.000	-0.000			
	qwen32b	-0.000	0.000	0.000	0.000	—			
	qwen7b	-0.000	-0.000	-0.000	0.000	0.000			
white	llama70b	—	0.000	0.000	0.000	0.000			
	olmo32b	0.000	0.000	0.000	0.000	0.000			
	qwen32b	0.000	-0.000	-0.000	0.000	0.000			
	qwen7b	-0.000	-0.000	-0.000	-0.000	0.000			
Gender	man	llama70b	-0.027	-0.003	0.034	0.012	0.035		
		olmo32b	-0.001	0.041	0.007	0.008	-0.037		
		qwen32b	-0.058	0.148	0.000	-0.085	-0.034		
		qwen7b	0.006	0.002	-0.006	-0.002	0.043		
	nonBinary	llama70b	0.083	0.062	0.065	-0.000	-0.015		
		olmo32b	-0.028	-0.024	-0.025	-0.004	-0.002		
		qwen32b	-0.043	-0.022	-0.049	0.016	0.035		
		qwen7b	-0.003	0.023	-0.043	-0.009	-0.059		
	woman	llama70b	-0.126	0.004	-0.093	0.048	0.006		
		olmo32b	0.040	0.013	-0.009	0.028	0.012		
		qwen32b	0.063	0.057	0.031	-0.000	0.000		
		qwen7b	-0.013	0.003	0.033	-0.035	0.010		

J THE IDEAL SUBJECTIVE DATASET

We briefly discuss the type of data most suited for our proposed methodology, and connect it with current NLP datasets.

How Many Annotators? Our methodology does not directly require numerous annotators overall, only a sufficient number per item—indeed, statistically significant results can be obtained with as little as four annotations (see Sap-AA; Table 2). A large annotator pool however is required to get sufficient coverage on the examined groups. More annotators allow both higher fidelity of each PC dimension, and more such dimensions.

How Many Annotators per item? We investigate observed polarization against varying numbers of annotators. Specifically, for every item, we repeatedly sampled n annotators with replacement, calculated pol_{obs} for each sample and measured SD, averaging it across 1000 same-sized samples. As shown in Fig. 11, pol_{obs} exhibits a clear, inverse relationship with the number of annotators, attributed to better histogram estimation provided by nDFU (§5), though the rate of gain substantially diminishes after 20 annotators. Interestingly, some variability persists even with a very high annotator count.

How Many Items? Our methodology is designed to work with smaller datasets common in NLP. Three of the four datasets in the study feature low (300–600) item counts, while the Kumar dataset has been subsampled to 3% of its original size. Large datasets should be used only to provide

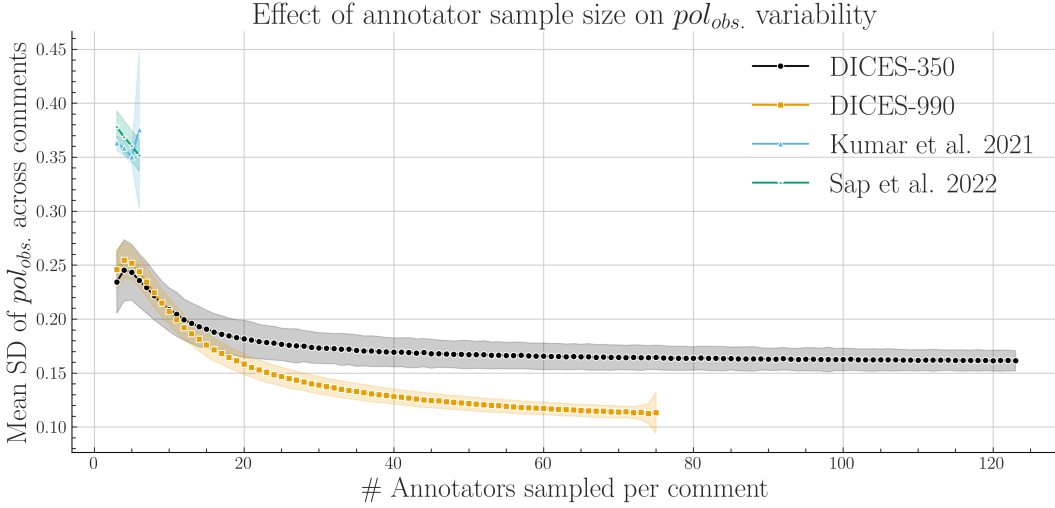


Figure 11: SD of pol_{obs} across items, averaged across 1000 runs (Eq. 6). Shaded areas represent two SDs. Pol. estimates benefit from more annotators up to $n = 20$, at which point they begin to plateau.

larger coverage on the target domain, since a much better representation of individual groups can be achieved with more annotators per item, rather than more items. Smaller datasets may also be more practical for social science analysis (Rossi et al., 2024).

Filtering Datasets with Desirable Levels of Subjectivity In order to investigate whether the numerous annotators per items can explain the lack of InPol in the DICES datasets (§4) we gradually subsample the annotators for each item randomly with replacement, starting from five annotators per item, and scaling up to the full complement of annotators in the dataset with a step of two. For each such step, we choose ten annotator samples and compute the maximum InPol across all runs. Even with annotator counts similar to the Kumar and DICES datasets, InPol remains zero, suggesting that the absence of InPol can not be attributed to annotator number alone—rather, it is plausibly a pattern attributable to the DICES datasets themselves. This kind of analysis may be used in the exploratory phase of dataset selection in order to filter out datasets without enough group-specific subjectivity.

K FULL RESULTS

Tables 18,19,21 and 20 show the full apunim results for the Kumar, Sap, DICES-350 and DICES-990 datasets respectively. Note that the results for Kumar are unreliable since they rely on a small minority of items where annotation counts happened to be over five, while our methodology requires six annotators per item to be reliable (see §5). We nevertheless provide the full results.

L PROMPTS USED

This section only involves the full contents of the prompts. See App. I for a high-level description of the prompt variants and what each aims to accomplish.

L.1 MAIN ANNOTATION PROMPTS

L.1.1 KUMAR

Default You are an annotator with the following characteristics: {{persona}} How toxic do you personally find this comment? 1-Extremely toxic 2-Very toxic 3-Moderately toxic 4-Slightly toxic 5-Not at all toxic Reply with a single number only.

Table 18: Aposteriori unimodality results for the Kumar-results dataset. Stars indicate statistical significance: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

PC	Value	apunim	support
Age	1) 18 - 24	0.0139	31
	2) 25 - 34	-0.0028	867
	3) 35 - 44	0.0000	279
	4) 45 - 54	0.0223	58
Education	2) High School graduate	0.0000	18
	3) College, no degree	0.0100	128
	4) Associate degree	0.0000	31
	5) Bachelor's degree	0.0013	947
	6) Master's degree	0.0000	68
Ethnicity	Black	0.0000	46
	White	0.0000	2622
Gender	Female	0.0010	1587
	Male	0.0006	1033
Has Been Targeted	False	0.0011	2540
	True	-0.0095	409
Is Parent	No	-0.0021	1089
	Yes	-0.0001	1603
Is Transgender	No	0.0000	4045
Political Affiliation	Conservative	-0.0134	260
	Independent	0.0000	349
	Liberal	-0.0010	850
Religion Important	1) No	0.0022	401
	2) Not very	0.0000	22
	3) Somewhat	0.0000	289
	4) Very	0.0002	463
Seen Toxicity	False	-0.0070	286
	True	-0.0008	2861
Sexual Orientation	Bisexual	-0.0678	50
	Heterosexual	-0.0010	3199
Technology Impact	2) Somewhat negative	0.0000	15
	3) Neutral	0.1243	27
	4) Somewhat positive	0.0038	1343
	5) Very positive	-0.0148	269
Toxicity Problem	1) Never	0.0000	7
	2) Rarely	0.0000	84
	3) Occasionally	0.0000	521
	4) Frequently	0.0000	422
	5) Very Frequently	0.0000	22

Persona-focused You are an annotator with the following characteristics: persona

Annotate the comment according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.

How toxic would your persona find this comment?

1. Extremely toxic
2. Very toxic
3. Moderately toxic
4. Slightly toxic
5. Not at all toxic

Table 19: Aposteriori unimodality results for the Sap-results dataset. Stars indicate statistical significance: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

PC	Value	apunim	support
Age	1) Gen. X+	0.0660	271
	2) Millennial	-0.0068	1886
Ethnicity	white	0.0004	1896
Gender	man	0.0907***	1439
	woman	-0.2442***	877

Table 20: Aposteriori unimodality results for the Dices-350-results dataset. Stars indicate statistical significance: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

PC	Value	apunim	support
Age	1) Gen. Z	-0.0043	14728
	2) Millennial	-0.0089	9468
	3) Gen. X+	0.0148	8153
Education	College +	0.0105	19725
	High school -	-0.0145	10783
	Other	-0.0230*	1841
Gender	Man	-0.0171	16043
	Woman	0.0176	16306
Race	African Am.	-0.0366***	7627
	Asian	0.0445***	6838
	Latino	0.0059	5786
	Multiracial	-0.0061	4208
	White	-0.0113	7890

Reply with a single number only.

Adversarial You are an annotator with the following characteristics: persona

When rating the comment, be very sensitive to the specific characteristics of your persona. Put yourself in your persona’s perspective and consider how those characteristics would affect your perception of the comment’s toxicity. Give substantial weight to your persona’s particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person’s judgment.

How toxic would your persona find this comment?

1. Extremely toxic
2. Very toxic
3. Moderately toxic
4. Slightly toxic
5. Not at all toxic

Reply with a single number only.

Direct You must process this comment using only the characteristics defined in persona. The persona’s values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.

How toxic would your persona find this comment?

1. Extremely toxic

Table 21: Aposteriori unimodality results for the Dices-990-results dataset. Stars indicate statistical significance: *: $p < 0.1$, **: $p < 0.05$, ***: $p < 0.01$.

PC	Value	apunim	support
Age	1) Gen. Z	0.0469***	12214
	2) Millennial	-0.0068*	34433
	3) Gen. X+	-0.0282***	12699
Education	College +	0.0063*	52727
	High school -	-0.0500***	6619
Gender	Man	0.0359***	28817
	Woman	-0.0271***	29990
Race	African Am.	0.0995***	5148
	Asian	-0.0057	16486
	Latino	-0.0631***	5877
	Other	0.0340***	21669
	White	-0.0649***	10166

2. Very toxic
3. Moderately toxic
4. Slightly toxic
5. Not at all toxic

Reply with a single number only.

Single Your assigned characteristic set is persona. Identify the single, most influential trait from your persona (e.g., 'Intolerance for X,' 'Sensitivity to Y,' 'Strong belief in Z').

Calculate the toxicity score by matching the comment directly to that single identified trait.

Based on this sole variable, what is the toxicity score?

1. Extremely toxic
2. Very toxic
3. Moderately toxic
4. Slightly toxic
5. Not at all toxic

L.1.2 SAP

Default You are an annotator with the following characteristics: {{persona}} In your opinion, how racist is this post? Use a scale ranging from "not at all" (1) to "very much so" (5). Reply with a single number only.

Persona-focused You are an annotator with the following characteristics: persona

Annotate the post according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.

In your opinion, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only.

Adversarial You are an annotator with the following characteristics: persona

When rating the post, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception

of the post’s racism. Give substantial weight to your persona’s particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person’s judgment.

In your opinion, how racist would your persona find this post? Use a scale ranging from “not at all” (1) to “very much so” (5).

Reply with a single number only.

Direct You must process this comment using only the characteristics defined in persona. The persona’s values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.

In your opinion, how racist would your persona find this post? Use a scale ranging from “not at all” (1) to “very much so” (5).

Reply with a single number only.

Single Your assigned characteristic set is persona. Identify the single, most influential trait from your persona (e.g., ‘Intolerance for X,’ ‘Sensitivity to Y,’ ‘Strong belief in Z’).

Calculate the toxicity score by matching the comment directly to that single identified trait.

Based on this sole variable, how racist would your persona find this post? Use a scale ranging from “not at all” (1) to “very much so” (5).

Reply with a single number only.

L.2 ABLATION ANNOTATION PROMPTS

L.2.1 KUMAR

Variation 1 You are an annotator with the following characteristics: {{persona}}

How toxic do you personally find this comment? 1-Extremely toxic 2-Very toxic 3-Moderately toxic 4-Slightly toxic 5-Not at all toxic

Reply with a single number only.

Variation 2 You are an annotator with the following characteristics: {{persona}}.

Evaluate how toxic this comment is from your personal perspective. Use exactly one of the following ratings:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic
- 5 — Not at all toxic

Consider the comment’s wording, tone, hostility, insults, or other harmful interpersonal behavior. Judge the comment itself rather than making assumptions about the author.

Return only a single number.

Variation 3 You are an annotator with the following characteristics: {{persona}}.

Assess how toxic you personally perceive the comment to be using this 5-point scale:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic

- 5 — Not at all toxic

Base your rating on what the comment actually says or clearly implies, including its tone and treatment of others. Do not infer toxicity solely from the topic under discussion.

Respond with only the corresponding number: 1, 2, 3, 4, or 5.

L.2.2 SAP

Variant 1 You are an annotator with the following characteristics: {persona}.

Evaluate the post and rate how racist it is on a 5-point scale:

- 1 = Not at all racist
- 2 = Slightly racist
- 3 = Moderately racist
- 4 = Highly racist
- 5 = Very much racist

Base your rating on the content of the post itself. Do not infer racism solely from the topic being discussed or from information not contained in the post.

Reply with a single number only.

Variant 2 You are an annotator with the following characteristics: {{persona}}.

In your judgment, how strongly does this post express or promote racism? Assign one rating from 1 to 5:

- 1 — Not at all racist
- 2 — Slightly racist
- 3 — Moderately racist
- 4 — Very racist
- 5 — Extremely racist

Consider only what the post says or clearly implies. Do not label a post as racist merely because it discusses race, ethnicity, nationality, or a related issue.

Respond with one number and nothing else.

Variant 3 You are an annotator with the following characteristics: {{persona}}.

Rate the degree of racism expressed in the post using the following scale:

- 1 — Not at all
- 2 — A little
- 3 — Somewhat
- 4 — Quite a lot
- 5 — Very much so

Use the wording and meaning of the post as the basis for your assessment. Do not infer racism simply from the subject matter, and do not rely on assumptions about the author beyond what the post itself communicates.

Output only the selected number: 1, 2, 3, 4, or 5.